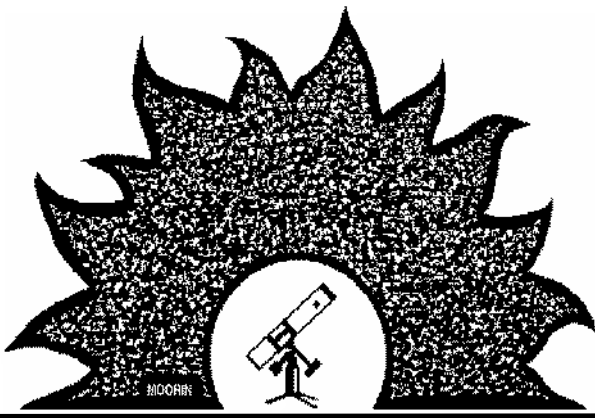




PAStimes

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Meeting of the Minds May 15

By Terri, Event Coordinator

At the Meeting of the Minds (MOM's) on May 15, 2014, we had 2 guest speakers lined up and bunch of topics to discuss. This was a great meeting with a wonderful turnout. We gathered in room K-115 as the Main Campus was in lockdown. Many thanks to all who brought snacks. Bruce provided the case of bottled water. Prior to beginning, and upon congratulating Rodney Fong for taking the position of Meeting Host, Bob Senzer presented the PAS Sign In Board with a new pen. This is a special Space Pen. That means it can write in any position, including upside down. Many thanks to Bob for donating this pen to PAS for the Meeting Sign In Board. Rodney, as new PAS Meeting Host, gets to enjoy use of the Pen while he has his position. Here's what the pen says on the packing... "Writes at any angle, in extreme temperatures (-30 Deg to +250 deg F), upside down, over grease and has a lifetime guarantee. Fisher Space Pen, Proudly made in the USA." It also says it will write underwater, & in zero gravity."

The Agenda opened up with a discussion about how we should upgrade the PAS website. Eric lead the discussion. We wished that Chris Johnson (PAS Webmaster) could have attended, so he could hear the concerns of the user-ness of the site, and maybe, if it were possible, help to make changes to accommodate the requests. From what I know about the software we are using, it is a set form and changes are not easily made. But, Chris might know some way to get around some of the items. It was a good discussion.

We then turned to talking about adding a database or some way to populate a table with 5 years of Sun Rise and set times,

Moon Rise and set Times, times of Twilight and other important factors about knowing when to start a star party, etc. The topic turned to maybe having a sub domain that can hook to the PAS website and have a separate way to do the calendar that is easier on me adding events as well as on the visitor to the page to find out what events we have. And maybe color coding the events, such as public events would be one color and private events would be another color, and then there are Paid events.

Then we talked about a topic that Darlene wanted to discuss, but was absent for. She saw on Apr 3 after a PAS meeting at 9:45pm heading west on Grovers, a bright light in the sky that lasted 15-20 minutes. She wanted to discuss this and find out if anyone else witnessed this event. I asked around, but no one had seen it, so the discussion ended there. I did ask the membership to research it on line and send it through the listserv, if they found anything about it, so Darlene would have the answer. I haven't seen anything go through the list serv, as of writing this article.

Bob Senzer took the next topic, which was How to get younger people interested in PAS. Many thoughts were shared. We could use the Student union and libraries, and ask if we can host meetings there, for free, to draw in more public, was one of the ideas. This topic needs to be researched more, and discussed again at a future MOM's after everyone has had a chance to consider ways to accomplish this goal, and then come share those ideas with the group.

We then turned it over to Alfred who introduced the NEW PAS Asteroid Search Group (SFA SIG). Alfred and I set up 2 meetings for this group: June 12 & July 10.



Meeting of the Minds on May 15 was an informative meeting with many shared snacks. Taken by Terri Finch.



Bob Senzer did a fantastic presentation. Photo by Terri.

This is a SIG that would meet once a month to discuss findings and help in the search for NEO's by use of their computer. Alfred wants to get this SIG going, so if you are interested in such as SIG to help out with, contact Alfred. See the website calendar listing for the dates listed above to get Alfred's contact info.

Mike then announced a Holistic Dispensary that was giving away a bunch of

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Upcoming PAS Events for July & August

By Terri, Event Coordinator

July 3: Public Fireworks party at Mike's home in Carefree. Potluck is at 6:30pm. Sunset is 7:42, Dark is 8:45, with a Star Party after all the 4th of July excitement is done. RSVP is with Mike at Primefactory@q.com. Everyone welcome! Bring snacks to share, like ice cream or popsicles, for during the fireworks.

July 10: SFA SIG Meeting. PAS Members only. RSVP is with Alfred. Location: TBA, Time: 7pm to 9pm.

July 12: Lecture #2 of Mike's Summer Lecture Series. RSVP required with Mike. Topic: "Earth Shattering Evolution." Potluck at 5:30, Lecture at 7pm.

July 13: Bookmans FREE Telescope Workshop. Everyone Welcome. Meet in the front window of Bookmans at 19h Ave & Northern Ave in Phoenix from 3:30 to 5:30. Bring your telescope, accessories, something to write on and questions. RSVP is required with Terri at Events@pasaz.org. When you RSVP, include the make & model of your telescope.

July 26: Lecture #3 of Mike's Summer Lecture Series. RSVP required with Mike. Topic: "Planetary and Stellar Jets" with a Star Party after the lecture. Potluck at 5:30, Lecture at 7pm.

Aug 9: Lecture #4 of Mike's Summer Lecture Series. RSVP required with Mike.

Topic: "Galactic Jets and Cosmic Evolution" Potluck at 5:30, Lecture at 7pm.

Aug 10: Bookmans FREE Telescope Workshop. Everyone Welcome. Meet in the front window of Bookmans at 19th Ave & Northern Ave in Phoenix from 3:30 to 5:30. Bring your telescope, accessories, something to write on and questions. RSVP is required with Terri at Events@pasaz.org. When you RSVP, include the make & model of your telescope.

Aug 23: Lecture #5 of Mike's Summer Lecture Series. RSVP required with Mike. Topic: "Upcoming Astronomical Disasters" with a Star Party after the lecture. Potluck at 5:30, Lecture at 7pm. ***

Mike Makes Front Page

By Terri, Event Coordinator

Mike, PAS's very own Meteorite Man, has been seen at tons of events over the years. He is the highlight of most events, stealing away groups of people who would be looking through our telescopes during a star party, but instead are mesmerized by Mike & his story of the meteorites he brings to those events. Schools have requested "Just Mike" to be at their events! He is a hot, in demand, item that only PAS has!

Well, he made it on the front page of the Puma Press, dated Apr 2014. Those of us in Astronomy are not pleased with the title on this page, but the fact that Mike got an article written up about him, is awesome! The title of this page is "STEAM event features night of astrology, mythology". For those of you who don't know... Astrology is not a science. The American Astronomical Society has this paragraph as the difference between Astronomy & Astrology, found on this page: <https://aas.org/faq/whats-difference-between-astronomy-and-astrology>. "As-

tronomy is a science that studies everything outside of the earth's atmosphere, such as planets, stars, asteroids, galaxies; and the properties and relationships of those celestial bodies. Astronomers base their studies on research and observation. Astrology on the other hand, is the belief that the positioning of the stars and planets affect the way events occur on earth. If you're interested in the solar system and the planets, other celestial objects like asteroids and comets, other galaxies and the rest of the universe, what makes up space, and the possibility of alien life or space travel, Astronomy is the field you're considering."

It was very nice of the Puma Press to put Mike on the front page of their newspaper. The article, except for it's title, is very good and all about the Black Mountain Campus star parties PAS hosts. I hope, in the future, we can stick to the Science of Astronomy, rather than telling the public that what PAS does is Astrology.



Mike deserves front page of every newspaper that comes to one of our events! Thank you, Puma Press, for giving Mike the recognition he deserves for all his hard work sharing his love of Meteorites with everyone!

Phoenix Monsoon Season

By Terri, Event Coordinator

A quick reminder about the Phoenix Monsoon Season: I wish to remind everyone that the Monsoon Season is happening now. Officially, it is from June 15 to September 30, this year.

What does this mean for you? It means that, if an RSVP is requested or required to attend a PAS event, it is very much to your advantage in saving driving time, and gas to get to an event, that may be canceled due to

the Monsoons, to be sure to RSVP your attendance so that I can get the word to you before you make the trip. The Monsoons can hit Phoenix at a moments' notice. We could have a great event planned (for example, at Mike's home in Carefree) and because a storm comes to Phoenix, whether it is a Haboob (major dust storm) or a drastic, overly windy rain storm, the event then becomes canceled. So, be sure you take the time to RSVP where requested. A simple

Email saying you are attending, could save you time and gas, if we have to cancel. Everyone attending should RSVP every time, when requested to do so, but not everyone does. This is my request, to you, to save you the hassles of getting to a canceled event, RSVP, every time, especially during the Summer months. Thank you for considering this request. Let's all have a great, clear night, starry star party & other events this Summer! ***

New PAS SIG!

By Alfred Fantegrossi

(Note from Terri: What is a SIG? A SIG is a Special Interest Group. Within the topic of Astronomy. There are so many subtopics that PAS Members can branch out as smaller groups under PAS. Ideas would be a Field Trip SIG, where by the members of this group would set up & arrange Field Trips for PAS Members to attend, or a Meteorite SIG, where by Mike might run this SIG & have meetings and field trips to different places that have Meteorites, or chase Meteorites when a large Meteor streaks across the sky & then falls to the ground. To start a new SIG, email Terri with the idea. With this in mind... Alfred writes the following for his new SIG.)

An article appeared in the June 2014 issue of "Sky & Telescope" (pg. 27) concerning an outreach program for high school & colleges as part of the International Astronomical Search Collaboration

(IASC). It is a group of professional astronomers, amateurs & teachers who established this free, web-based asteroid discovery program out of Abilene, Texas. Mostly conducted in schools and colleges, it is also available to Astronomy Society's such as ours.

I have contacted the director of this group, a Patrick Miller, from Hardin-Simmons Univ., and he has invited our participation in analyzing several sets of Astrophotographs from various large telescope surveys of Kuiper Belt/ It may also involve NEO's to identify or confirm tracking measurements for.

Prime-focus telescopes involved in this effort are as follows: 1) University of Hawaii's Pan-STARRS 1.8M PSI (spring & fall campaigns), 2) ARI (Astronomical research Institute, from Westfield Illinois 50" reflector prime focus

telescope- year-round photos contributed. 3) 0.8meter Schulman Telescope at Mt.Lemmon SkyCenter

Currently, our group will be entered into this Aug and October's distribution of survey photos to scan. Using a special computer analysis program, we will participate by scanning the photos, for asteroids and comparing them with known object databases.

We are looking for any members who would be interested in such a project. Our analysis will be time dependant, as these photos will be available to us within a short period of time (? hours-days) after having been taken by the various survey telescopes. Please Let me know your thoughts ,questions, ideas about participating in this Special Interest Group. Thanks you. Al Fantegrossi ***

September PAS Meeting Planning

By Terri, Event Coordinator

At the September PAS Meeting, as we have done for several years now, we will be having Members' night. What this means is that the PAS Members will do presentations. But this is also open to anyone wanting to present something to the PAS Members, without being a PAS member. So, if you have something you'd like to present to the club, anything about Astronomy (or related topics), you need to get in touch with me by Aug 15 to be scheduled to do your presentation. I will need to know the title of your presentation, the length of the presentation and you need to submit, for the September newsletter, a brief teaser paragraph about your presentation. If you miss the deadline for the newsletter, you

can still sign up to do a presentation, but you won't be announced in the newsletter.

At the moment, our first presenter of the evening will be Leah who will have 2 mini presentations for us via Skype. At the Meeting of the Minds this past May, we had her doing a presentation and 10pm came and the computer shut down. Everyone took that to mean the meeting was over, so we didn't get to see the end of the presentation. So, Leah will be repeating that presentation, as many PAS members who were in attendance, mentioned they'd like to hear the "rest of the story." Then, she will present another really good presentation. Both of these will be short but well worth your attendance.

After Leah, nothing is planned, at this time. Now, if I get too many presentations for the Sept 4 PAS meeting, we have a Meeting of the Minds for Sept 11th, the following Thursday, for which we can fill with presentations as well. Let's make September a great month to hear some awesome presentations. Get your data to me sooner than later, so we can discuss the details and put you on the schedule. Thanks so much for preparing such awesome presentations for the club! We open both the PAS meeting and the Meeting of the Minds up to Documentaries, Movies, Show & Tell and anything else anyone wants to share. See you there! ***

Meeting of the Minds May 15

(Continued from page 1)

monies. We had a discussion about this and decided we didn't want to be part of that kind of group. Even for free monies.

We then took a 10 minute bathroom break.

We then turned it over to Bob Senzer who did a presentation on "Alvan Clark & Sons." This presentation was very good, funny, and informative. I am looking forward to the next presentation by Bob, in which he is doing another historical account of someone famous. Possibly at the Sep-

tember 2014 PAS meeting. Bob's talk was great info and caused a lot of discussion.

We then hooked up Leah Sapir to Skype, with William's help and she began her presentation around 9:40pm. About 9:50pm a notice popped up on the screen that says at 10pm the computer is going to shut down. We let Leah continue and then the computer shut down, cutting off her presentation. That was somehow taken as a cue for everyone to leave for the night. It took us over 20 minutes to get reconnected, so we apologized to Leah and asked her to

do this presentation again at the September 2014 meeting. She will be first on the list to do a presentation about "Williamina Fleming." It is about 30 minutes long and if we had hung out just a little longer, we would have gotten the whole presentation after the computer shut down, 10 more minutes. So, plan to see this fine presentation at the September PAS meeting and a secondary one, also about the same length, and Leah is scheduled to be the first to present at that meeting, so she can get it all in.

Social Media SIG

By Terri, Event Coordinator

PAS is considering starting a Social Media Sig. What this would entail is having designated people, who belong to the Sig, receive emails from me, with updates to whatever Social Media they are maintaining for PAS.

So, you'd have to have a computer to do this, or computer access on a daily basis (although, you wouldn't be doing things for the SIG, daily. You'd be doing them as needed). You'd have to have a little spare time, and you'd have to be interested in tTwitter or Facebook or other forms of social media to announce upcoming events, update people on what's going on with PAS, send out cancel notices, etc. What started this conversation at Bookmans, June 8, was the idea that PAS should be part of Meet Up, which has a monthly fee. We would have to approve the monthly fee (through a PAS discussion at a Meeting of the Minds), but one person could post events to Meet up. They'd work through me, i'd send them the meet up info, they'd review it to make sure everything important is listed, such as

date, time, location of event, what we are doing, and then they'd post it to meet up and if a cancel notice was needed, they'd be in charge of getting that posted as well.

If you are interested in doing something like this, such as Tweeting for PAS, or other forms of Social Media, consider joining the new PAS SIG: "Social Media." We can use the help. The term of office would be 1 year at a time, as a Directorship (appointed, not voted in), so if you can do it this year, let's say, from Sept to May, but not next year, you could let someone else take the position, if needed, or if you really like what you are doing, you could continue as the Social Media Director. And we would have several Directorships to fill.

I would guess that when we have a social event coming up, it would be busy for a few days before the event, up to the event. And the rest of the time, there wouldn't be much to do. However, to start out, we would be posting all the upcoming events, as I get them.... such as Black Mountain

Campus, or the telescope workshops... all could have meet up possibilities.

Are you interested in helping PAS through the use of your computer? Then consider being part of this TEAM. Signing up doesn't mean you will have the position, right now. It means we will consider you for the position, and one person per Social Media venue will be chosen. Hopefully enough people want to help with this so we can cover many media venues.

The first goal of this SIG is to make a list of all the venues for Social Media, out there, besides Facebook, Twitter & Meet up. Then, we'd figure out how many we have, and how many Social Media Directors we have, and divide them up. The PAS Facebook page is currently maintained by William and me. We don't have a Twitter following, yet. And we don't have Meet Up.

This is an easy, at your computer, way to help promote PAS. Thanks for considering helping with this new SIG. Come be part of the Social Media TEAM. ***

For Sale: Telescope: 5" Celestron C5 Schmidt-Cassegrain

Focal Length FL = 1250mm at f/10, 5x24 spotting scope and bracket, 1.25" star diagonal and erecting prism, 1.25" Celestron 32mm plossl eyepiece, Piggyback platform (camera attachment rail), Thousand Oaks Glass Type 2 full aperture solar filter, Orion padded travel case for C5, Original manuals

Mount: Celestron CG-4 GEM tripod, mount and weights (new), Dual Axis CG-4 Motor Drives (R.A. and DEC), Hand con-

trol (for guiding, not slewing) 2X, 4X and 8X, Power Pack holds 4 D-cell batteries (up to 20 hours), Original R.A. and DEC hand control knobs, Polar axis finder scope for CG-4 equatorial mount, Tools: screw driver, and regular and hex wrenches, Original factory shipping box, Original manuals.

All for \$599 (see photo), local area only... Contact Alex at (602) 789-6556 or see me at a PASAZ meeting.***



Telescope Help Session at Mike's Apr 19

By Terri, Event Coordinator

This short notice event turned out well. The idea was that because I needed to set up some things on my telescope, we would ask Mike to lend us the use of his home to do this, and William & I didn't feel we needed the whole place to ourselves, so we opened it to anyone wanting to attend. And the secondary idea that came to me at the Bookmans Telescope Workshop of Apr 13, was that we had some new members who might want some telescope assistance. So, I created this event. Mike requested Ham and Pineapple Pizza, and so William and I brought 3 large pizzas. There were sides provided by Howard and Tuna and rolls provided by Bob. Rodney provided dessert, as always, and Angie brought Chorizo dip and chips. It wasn't a large dinner, but it fed everyone nicely. This was a PAS Members help session. So, with exception of Mike's Joe (who is currently not a member), there was no public in attendance.

So, in attendance from PAS was: Mike

Marron, the host of the party, Howard Moneta, Angie Lee, Rodney Fong, William & Terri Finch, Eric Steinberg, and Bob Senzer.

The evening began with the potluck. And they were so nice. William and I were running late by about an hour. So, when we arrived, Rodney rescued me from under the pile of pizza. We all went in and ate. Then, Howard and Rodney set up their scopes. Because I was suffering from a massive migraine in which I was not only in pain from, but also very drowsy and clumsy, I didn't want to take out my scope to do anything with it for fear I'd drop it. I had done dishes earlier that day and nearly broke a few, banging them in the sink unintended. Nothing broke, but I didn't want to chance that with my scope, so I was outside for a short time, visiting, then I went in and spent the rest of the night indoors to get away from the allergies, or whatever is causing all these migraine.

So, after dinner and a little viewing outside, we turned to movies. For a while there, several people watched some movie, then Eric had brought "Primer" of which I watched. Very interesting, hard to follow, especially with a migraine, and I will have to watch it again, sometime. I wish to apologize to everyone at this event for me not being able to be more involved, especially outdoors.

And I'd like to thank those who provided food for the potluck, and who attended with their scopes and for Mike, hosting the party. Thank you Mike!

Rodney writes: There were some thin clouds early on, but it became clearer as the night progressed. Saw Jupiter, Mars and later Saturn, as well as Orion's Nebula and Big Dipper's Double Stars. Howard's more capable scope caught various clusters, nebulae & globulars. It was a nice star-filled night. ***

Desert Botanical Gardens Apr 26

By Various Attendees

Six scopes were asked to attend this event. PAS Provided Kevin Witts, Sam Insana, Rick Cunningham, Earl DeLong, Alex Vrenios and Howard Moneta. Amanda Ellis was our contact at DBG.

Earl writes: Sam, Alex, Kevin, Rick, Howard and I had a wonderful night at the Desert Botanical Gardens on April 26, 2014. We provided a star party for the guests of the major fundraising event. It ended up being a beautiful spring evening despite the storm passing through during the day. Only during the end of the event were scattered clouds a problem.

We arrived between 7:30pm and 7:45 pm. The carts arrived and we began transporting our equipment to the wildflower garden. We then moved our vehicles to the parking lot and began setting up our equipment. The guests were done with dinner around 9:30pm and began strolling the grounds and looking through our telescopes.

I started with M44 the Beehive cluster and then moved to the beautiful double star Iota Cancri. Howard stayed on Mars and I let him borrow my 9mm Nagler eyepiece which gave a nice view. Alex stayed on Jupiter. Kevin showed several objects with Saturn getting the most ohhhhs. Sam

showed the galaxies M81/M82 and Rick was on the double star Castor in Gemini at one time.

We had a large crowd throughout the event. Everyone was appreciative and interested in what they were observing. It was nice that the DBG staff had a slide presentation of astronomical pictures playing on a large screen near our set up area. This did not hamper our observing at all.

I would definitely return to the DBG for another event. The staff were very helpful and the public were pleasant. Everyone enjoyed the evening.

Alex writes: The Desert Botanical Gardens held a fundraiser Saturday evening and we were an extra treat for the attendees. After the dinner and all the speeches, some of the Garden's most important benefactors came out to find a half dozen scopes trained on Mars, Jupiter and (later) Saturn. Some of us tried to show M13 but there was too much ambient light for it to show up as anything but a faint patch. We had better success with the star Castor in Gemini, a blue/red double, but we mainly focused on the three bright planets in our night sky this month. About two hundred people stopped by our setup area and at least half of them

took a look through my telescope. Some were well versed in the planets and their moons, but most were eager to learn about them and their historical relevance. Many saw the moons of Jupiter and rings of Saturn for the very first time.

Sam writes: On Saturday night, April 26th, 6 members of the PAS Star Tours group attended a star party at Desert Botanical Gardens. Kevin, Howard, Alex, Rick, Earl and Sam participated. From 9:15 pm until 11:15 pm we showed Jupiter, Mars, Saturn, M 44 beehive cluster, Cancer iota double star, and Mizar double star. There was too much ambient light from the Chihuly exhibit and other lights to show any galaxies, globular clusters, nebulae or other of the usual deep sky objects. Over 200 people looked through our 6 telescopes and they were very excited, especially to see the rings of Saturn. The wind was less than 5 mph and not noticeable, but we had about 20-25 % cloud cover. Luckily the clouds were like cotton balls with space in between so we could still show the objects. Much better than a thin cloudy sky covering everything. Our DBG contact Amanda was quite helpful getting the motorized carts for us to load and unload our equipment. She also showed us where exactly to set up, and

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The Planets This Month

By Leah Sapir



Jupiter is fading into the sunset this month. It will still be visible in evening twilight in the first few days of July, but after that it will sink down in the west, and join us again as a morning star towards the end of August.

Mars and Saturn are still putting on a good show in the evening hours. Mars is in the southwest after sunset, and sets around midnight at the beginning of July, 11 pm at the end of July, and 10 pm at the end of August. At its peak, in April, Mars was around 60 million miles away from us, with a visible diameter of 15 arc seconds, and shining at a magnitude of -1.4 , about as bright as Sirius. Now it is about 90 million miles away, with a diameter of only 9 arc seconds and a magnitude of zero, like Arcturus. By the end of August it will be almost 130 million miles away. Its diameter will be only 7 arc seconds and its magnitude around 0.65. That's only a little brighter than Antares, which incidentally will be quite close to it in the sky. At the beginning of July, Mars will be in Virgo, but during August it will be moving through Libra, and at the end of September it will have a close encounter with Antares. That will be quite a photo op!

Saturn is in the south after sunset, and will be up till around 2 am at the beginning of July, midnight at the end of July, and 10 pm at the end of August. Mars and Saturn

are about 30 degrees apart at the beginning of July, but as Mars moves eastward it gradually moves towards Saturn and passes it on August 22 at a distance of about 3.5 degrees.

Mercury starts July as a morning star, visible only in morning twilight. Its earliest rising is around 4 am in mid-July. After that Mercury fades into the sunrise, and reappears in the evening twilight at the end of August. It won't be very visible in either case.

Neptune and Uranus are evening stars, with Neptune reaching opposition on August 29. Neptune rises around 11 pm at the beginning of July, 9 pm at the end of July, and 7 pm at the end of August. Uranus rises about two hours later, and reaches opposition in October.

Venus is still a morning star, rising around 3:30 am in July, 4 am at the beginning of August, and 5 am at the end of August.

New moons are scheduled for July 26 and August 25; full moons on July 12 and August 10.

The first-quarter Moon will accompany Saturn across the sky from south to west on July 7-8, from sunset to 2 am. On July 21 the third-quarter Moon will be near the Pleiades in the east from 2 am to dawn, and on July 22 it will be near the Hyades from 3

am to dawn. The crescent Moon will be close to Venus in the east before sunrise on July 24.

From sunset to 11 pm, the first-quarter Moon will join Mars in the west on August 3, and Saturn on August 4. Then on August 18 the third-quarter Moon will accompany the Hyades rising in the east from 1 am to dawn. On the same morning, August 18, we will see a conjunction of Jupiter, Venus and the Beehive Cluster low in the east before sunrise. All three objects will also be relatively close to each other for a few days before and after; and on the morning of August 23, the crescent Moon will join the party.

In the west after sunset, Mars passes by Saturn on August 22. They will be relatively close for a few days before, and will stay together till the end of the month. On August 31, the crescent Moon joins the group as well.

The Perseid meteor shower will peak on the night of August 11-12, but of course the full moon will interfere. Perseid meteors can be seen from around July 13 to August 26, so it will be better to watch for them around the times of the new moon.

Join us next time when we continue to explore the Arizona sky. And till then, wishing you clear skies and happy observing!

Ron's Planetarium Opening Day June 14

By Alfred Fantegrossi

Ron Walker's Planetarium saw First Light on June 14th (following Friday the 13th...) I attended the 5pm showing on a beautiful, if not hot Cave Creek Day, along with about a dozen other guests, including a family with young children.

He told me that this had been his dream since 1954 and that his architectural model was a geodesic dome covered in similar material that was part of the space vehicles. It is highly reflective, so that the interior of the planetarium would be easier to keep cool.

The planetarium "projector" was a used projector from Texas, that he bid for and won (for a sum of 1 cent on the dollar, of purchase price... a bid he won by a mere \$200 over next highest bid)

The show itself started with him dimming the lights in the planetarium slowly to allow the guest's eyes to dark adapt..and stated that the sky that we would see would reflect those seen for the season and location of the show itself. The "Theater of the Stars" began.

The show itself was fairly interesting, mentioning how 500 yrs ago, man might have theorized that Earth was center of



Inside Ron Walker's newly built Planetarium in Carefree. Photo taken by Ron & provided to PAS via Alfred

Ron's Planetarium Opening Day June 14

universe, but that certain movements of the planets conflicted with that theory, and how Sun-centered picture made better sense of predicting planet movements.. Then to the constellations and their mythological origins..

But Ron had bigger plans for his new Planetarium, with modifications to the physical plans as well as additional pro-

graming to the shows in the future. 1) a light-trap entrance vestibule, for "latecomers", to protect light-adapted guests already enjoying the show. 2) Program additions- to include demonstrations of the Solstices, further constellations and meteor showers etc. 3) He is still awaiting final inspection of his Planetarium, but everyone present seemed to enjoy the show immensely, and

we could tell by the smile on Ron's face, that he was pleased that all his hard work had been justified.

PS. (on a note of recommendation, I did think that the A/C did compete with his audio during the presentation, but we were comfortably cool inside...on a 100deg evening....so who's complaining?) Thanks for inviting us to your grand opening! ***

PVCC/PAS Meeting Room

By Terri, Event Coordinator

Many thanks to Jenny Weitz, our Liaison at PVCC, for getting us another fine year of trading the PVCC rooms for star parties for our events & meetings. For the

Telescope Workshops, we are in G-147, which is the room by the Telescope Dome. And for our General Meetings (1st Thursday of the month - in most cases) and our

Meeting of the Minds (Business Meetings) we are in LS-205 for the 2014-2015 school season. We will see you there! ***

Black Mountain Campus Apr 21

By Various Attendees

In attendance at this event from PAS was: Sam Insana with his scope, Kevin Witts with his scope, Alex & Diane Vrenios with his scope, Eric Steinberg with his scope, Howard Moneta with his scope, Mike Marron with his meteorites, Roger Anzini with his scope and Michelle, Earl DeLong with his scope, William & Terri Finch with Handouts, Rodney Fong with his scope, Darlene Ahlefeld (with her uncle who visiting from Florida), Bob Ewing with the PVCC scope (12" Light Bridge), & Dick Newell.

Sam writes: I showed several objects with my 8 inch Newtonian reflector on Monday night at PVCC Blk Mtn: Jupiter, Orion Nebula, Saturn, M13, M81 and M82.

I tried to find Comet C2012 K1 Pans-tarrs and had no luck. About 30 people looked through my scope, including 5 people from my golf group. There were several college students who needed a lot of information for their astronomy class and they took notes of the telescope specifications and detailed information on the celestial objects they were observing. I noticed over 100 people in attendance at the star party and everyone seemed to enjoy themselves. The 8 weather forecasters accurately predicted cloud cover of less than 20%, making for a good night of observing. Although the International Space Station made a dramatic appearance at 10 minutes before 9 pm, I did not see any of the Lyrid meteor shower. Mike made up for this by showing his excellent meteorite collection to the crowd.

Alex writes: PAS held a public star

party at PVCC's Black Mountain Campus Monday night, April 21st. I brought my 5" SCT and had a much better time of it, not having to deal with the unusual GoTo requirements of my new 6". There were about 60 people in attendance, students and locals, in PVCC's partially darkened parking lot. I gave laser pointer tours again, as needed, and showed off the Orion Nebula, the Pleiades, Jupiter and its moons, and brought attention to Ceres, an asteroid that was visible naked-eye at this very good, if not truly "dark" site. The atmosphere seemed quite steady, allowing us to see detail on the surface of Jupiter, and the glowing gas and dust cloud in M42, making for some interesting discussion among those seeing it for the first time.

Terri writes: William & I attended with only handouts at this event. William didn't have the next day off from work, so we wanted to be able to get home at a decent time to get him up for work the next morning at 4am. So, when 10pm came, we packed up and left, almost that fast.

I estimate over 100 people attended this event, not counting PAS Members. Jodi Sullens joined us. She attended a previous BMC event (Feb 27) with her scope. This time, she came as an observer. I enjoyed her company and our discussions about her telescope and astronomy.

The skies were amazingly clear. The public seemed to totally enjoy this event. The temperature was perfect, no jacket needed. At 6pm when William & I arrived, it was 96 degrees and hot. Rodney was

already there and so the three of us made a trip to the bathroom, and on the way back, we stopped in at the Security office so that I could meet the new Security person. Loreta Mondragon introduced me to Mike (I will miss Kevin, our previous Security dude). Mike and I chatted for a bit, and then William, Rodney and I went outside to set up the Handout table. Many thanks to Rodney for his help through the night and for offering me his chair when my back started hurting.

Many PAS "Who are we" pages were handed out. 5 people signed up to receive notices about upcoming public events. Many students were there from SCC and PVCC (both campuses).

A Guest that I invited, Brenda Hamblin & her daughter, Dione, attended. They were very amazed. They stayed until ISS went over at 8:50pm. We watched ISS. That was very nice. I haven't seen ISS flyover in a few years.

I walked around, at one point, when the public wasn't by the Handout table, and talked with all my Team members to find out what they were showing (for this review). Through Eric's scope, I saw Saturn. Eric and I counted 7 moons. Roger also counted 7 moons in his scope. Bob Ewing and students from his class were trying to photograph Jupiter and find out why we couldn't see all 4 of the Galilean Moons. They were using their phones. As they got involved in trying to capture Jupiter in their phone cameras, I had seen Jupiter and moved on to other scopes.

(Continued on page 22)

PAS Social Jan 18

By Terri, Event Coordinator

A very big Thank you to Chris & Tiffany Johnson for hosting this year's PAS Social. And many thanks to those who competed in the newest event at the Social, which was the Social Cook Off. And thank you goes to all the attendees of this party, who made this party a very fun event to attend.

So, the event started with the set up of the tables for enjoying dinner. Ed, Bette and Bruce Wurst showed up just in time to set up 2 long tables and a few chairs for dinner. The main dining table was already there and Frank & Sam Insana brought a card table for added eating space. While everyone was arriving and adding their food to the collection in the kitchen, I was handing out White Dwarf Gift Exchange Pink Tickets and Door Prize Blue Tickets.

In attendance and bringing the following foods for the potluck were: Rick Cunningham with scalloped potatoes, William Finch had his Queso Dip and chips. Along with Terri Finch they provided Chocolate Peanut Butter Stuff, a Rotisserie Chicken and a tub of Potato Salad, Ed Wurst ordered 2 pizzas to be delivered to the Social around the same time he was arriving. Ora & Eric Steinberg brought Oatmeal Raisin Cookies. Mike Marron attended, but I didn't see what he brought for food. Sam & Frank Insana brought the Meatballs & Pasta in Marinara Sauce, Darlene Ahlefeld brought Sugar Cookies, Matt Kohl and his nephew Cole brought White Chili & Green Stuff, Darren Johnson brought 2 large pans of lasagna and several tubs of parfait, Jerry Belcher brought Hot Wings & Pizza, Chris & Tiffany made Redneck Sushi and Rodney Fong brought dessert. Terry Dancer and fiancée Laura came after the food. Steve Palmer showed up with pizza to share and stayed for only about 1.5 hours.

During dinner, there was a white board with the list of foods we were judging. Many thanks to Matt Kohl for helping me hang and set up the signs by the foods that were in the Cook Off. The rules for this competition were 1) Don't vote for the food you brought, 2) Don't vote for anything you didn't eat, and 3) Circle one winner from each category. The categories were: Main Dish, Side Dish and Desserts. Originally the Main & Side dishes were to be combined into one category, but one of the PAS members said she was bringing a prize for the Side Dish category, then due to her daughter

becoming ill that morning, she didn't make it to the event. Thus, the Side Dish competition didn't have a designated prize when it came time to award the prizes. And 2 of the dishes in the competition were also hers, and were not able to be voted on. So, as dinner concluded, we did the voting. In the Cook Off but not winning this year was Terri's Chocolate Peanut Butter Stuff, Cole's Green Stuff, Chris's Redneck Sushi and Darlene's Sugar Cookies. The winners of this year's Cook Off were: Main dish winner with 7 votes was the Pasta & Meatballs in Marinara Sauce made by Sam Insana. Sam was awarded the Astronaut Cookbook provided by PAS. The Side dish was the Queso Dip from Hell with 5 votes, made by William Finch who chose a door prize of Year in Space Desk Calendar. And the Dessert category had 7 votes for Ora Steinberg's Oatmeal Raisin Cookies and she was awarded the prize of a box of astronomy related snacks (of which I asked her to provide me a list of what was in her prize box: Here's the contents of my prize bag: Sun Chips, Star Wars mugs and cocoa mix, Cocoa Collection, Orbit gum, Starburst, Starlight mints, Angry Bird space fruit snacks, Space toothpaste, Milky Way bar) provided by Matt Kohl. The food, in general was all good, delicious! Thank you all for participating in the potluck!

Then we had the yearly Dart game. Many, many thanks to Ed and Bruce Wurst who set up and ran the dart game, and provided the darts and dart boards (the easels were brought by me). We did the darts on 4 boards this year for about 1.5 hours and Jerry Belcher provided 4 trophies. During the Dart game, there was a movie & lots of socializing indoors. The weather for this year's Dart game was so nice. We had a mini propane heater to huddle next to, but didn't need it as much as past year. Many thanks to all those who participated in the game. Participants were: Bruce Wurst, Ed Wurst, Sam Insana, Cole (nephew to Matt Kohl), Jacob Johnson (Chris's son), Jerry Belcher, Don Boyd, Bette Wurst, Rodney Fong, Ora Steinberg, Mike Marron, Frank Insana, Terri Finch, William Finch and Eric Steinberg. Part way through the game Jacob and Eric dropped from the game. There was a tie for 2nd place with a score of 6. So, we had a play off and the results were: Taking 4th place was Don Boyd with a score of 7 who got the 4th place trophy and, from the Door prize table, got the World Space Mu-

seum Cards. Taking 3rd Place was Rodney Fong with a score of 15, receiving the 3rd place trophy and a door prize of a Constellation puzzle. Taking 2nd place was Jerry Belcher with a score of 17 and a door prize of Hobbit Pez. First place was won by Bruce Wurst, who got a trophy and a dart related calendar provided by Jerry Belcher.

We then had the White Dwarf Gift Exchange game. It was really good this year, with 2 really great gift stealing, laughing, fun times during the game. Here is what everyone won during the game: Darlene won a Meteorite Dig Kit & a book "Backyard Night Sky," Jerry won the movie "Riddick" and a Saturn Puzzle, Terry Dancer won "Night Sky" observational tool kit, Terri won a book "50 ideas you really need to know - Universe" and a 2014 Year in Space Calendar, William won a gift certificate for Orion (which we will put to great use), Sam won a Constellation Mug which everyone was fighting over. Darren won a Laptop bag, a Calendar and an observing book, Tiffany won a 2014 Calendar and Dawn Mullan's Art, Cole won the "Dictionary of Astronomy" and an eyepiece case, Matt won a "Walking Dead" board game, Mike won a 2014 Calendar and the "Handy Space Astronomy Book," Jacob won "Mysteries of Space," Bette won an astronomy (means it has the ability to be Red as well as white) Flashlight and an eyepiece case. Bruce won an astronomy floor Mat with the Earth and words that say "You are here," and a singing / tweeting bird, and this is the other gift everyone fought over, Ed won an Einstein shirt and a book "Seeing Further."

Before we gave out the Door prizes and other awards for the evening, Bruce presented to Chris and Tiffany, a thank you card signed by all who attended this event. Then for the Main Door prizes, Mike Marron won a book "Utah Star Watch," Don Boyd won Star Trek Pez and Terri won Space Fact Cards. While clean up happened, Chris put on and provided 3D glasses to view "Star Trek - Into Darkness." Many thanks to everyone who provided door prizes. We had enough to go around and 2 extras this year. Many, many thanks. 2 door prizes were provided by Mike, and 3 were provided by President Bruce, and 2 were provided by Matt. We look forward to next year's event and your participation at it. ***



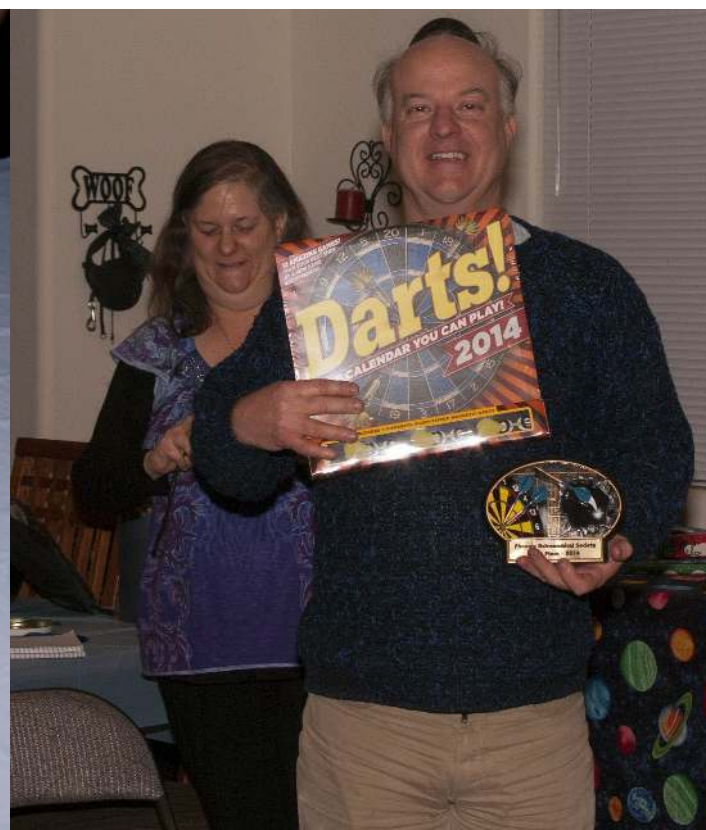
Tiffany is seen stealing a gift from Darren during the PAS White Dwarf Game. Photo by Don.



PAS Dart Game. The competition is rough this year. Photo taken by Terri Finch.



"Redneck Sushi" was Chris Johnson's entry in the PAS Cook Off at the 2014 PAS Social. Photo taken by Don Boyd.



Bruce Wurst is the 1st Place winner of the 2014 PAS Dart Game. Photo by Don Boyd.

PAS Is Better Than Cosmos!

By Stuart Samberg

I watch "Cosmos" and learn some things. When you, Mike, spent time with me at BMC explaining lots of things about Earth, it made me think how lucky we are to

have you. It was better than "Cosmos!"

An idea I had is to have your lecture series, excerpts from it, or other topics on your website. You could have a page and

call it "Meteorite Mike" Seriously, I think it would be awesome to have that kind of information on there. ***

Antennas Review Apr 28

By Eric Steinberg

Our trip this month to Antennas had a bit of a different twist – we were accompanied by two delightful and knowledgeable international guests: Mike Brightmore who is chairman of the astronomy society in Cleethorpes, England and his son-in-law Pierre-Alain of Montreal. They brought their 80mm refractor, tracking mount, binoculars and camera and followed me out to the site. It was something of a gamble since the winds were really strong when we arrived, too strong to set up the 22". Thankfully, the winds decreased after sunset sufficiently to allow observing and stayed that way until around 11pm when they came back redoubled ending my observing for the night. The skies themselves were some of the best I've seen with transparency a sparkling 10 and seeing (surprisingly) 7-8. Our guests, had never seen skies like that and were astounded. I don't know that I've ever seen two happier astronomers

(other than perhaps me the first time I went there), and was really gratified to be able to share the experience with them.

As far as observation I was able to get 27 more Herschel 400 objects (73 to go) before the wind shut things down. We also looked at Jupiter, Mars and Saturn – lots of clear detail in the wind lulls. Other objects were M51 (probably the best view of it I've seen in my scope), M104 and topping the evening, NGC 5139 – Omega Centauri, looking its usual spectacular self. Quote of the evening from Mike: "It's just wonderful to go to sleep with a head full of stars." I know just what

he means.

Terri adds: Photos for this event are in the PAS Photo Gallery at this link: http://www.pasaz.org/forums/gallery.php?g2_itemId=14256. ***



*Photo of Eric with his telescope at Antennas.
Photo taken by Mike Bridgemore.*

Rancho Gabriela Career Day May 6

By Terri, Event Coordinator

This year's event was set up like last years' as far as the location we set up. We arrived at about 7am at the school. We started to set up right outside the Gym under the roofed area. We set up a little differently this time. We took two tables to eat food at and put them to the back of the area. We took 1 table to put the PAS handouts on with the usual table cloth.

In attendance at this event was Jerry Belcher with a solar capable scope, Matt Kohl who did the PAS Handouts for me for the 6, 7, and 8th graders, William Finch who used our new PST at this event, Don Boyd with his 6" and a solar filter and I did Q & A with all the grades lower than 6th grade and a few kids from the upper grades. The idea was to get the PAS website out to as many kids as possible, and the flier that Matt was giving out, had that info on it, so we were trying not to duplicate giving the info out.

8am: we were set up and ready to go. It was not that hot at 8am but by noon, it was rather warm outside. We had a breeze with a few gusting winds. All was good until about 1pm when the cloud cover became 100% and in the distance, we could see a dust storm heading our way. So, about that time, I mentioned to our contact at the school, Julie Merrill, that we needed to pack

up because we lost the sun, the winds were too much for our smaller scopes, and the dust storm coming our way looked rather threatening.

Packing up, Don, William and I were last to leave and we were out of there around 2pm, but the winds were getting extreme by that time. And this was forecast on the 4:30am news ch 3 that morning. So, i knew it was coming.

It was a great event. We had many students come past our station, get handouts, win prizes, etc. The Fire department guys, just before they got a call, came over to look through the scopes. They were called away, then they returned, they weren't around for long before they were called away again. Many photos were taken. They will be put in the PAS Photo Gallery over the summer, so watch for them and Enjoy! We look forward to next years' event for Career Day at Rancho Gabriela. Many thanks to Matt for making the Career Day flier for this event. ***



Jerry Belcher is seen showing the Sun through his scope with a long line of Kids waiting to view it. And Don Boyd is in the background showing that line of students, the Sun. Photo by Terri Finch.



William Finch is showing the students of Rancho Gabriela School, the Sun through our new PST. Photo by Terri.

PAS Awards Ceremony Apr 24

By Various Attendees

Alex Vrenios writes: The last Thursday night in April is traditionally the Phoenix Astronomical Society's Awards Ceremony, and April 24, 2014 was no exception. "Wear your name tags," the announcement said, "because we're limiting attendance to current PAS members only!"

The plan was to have 2-to-3 slices of pizza for everyone, a table full of prizes for awardees to choose from and a few presentations to boot. Darren Johnson brought enough pizza to feed an army of astronomers. (Thanks Darren!) The pizza boxes were stacked two or three high and covered the entire back table.

The awards were presented to the winner of this year's Messier Marathon, to the winner of the "best costume" at the Halloween Party, to those who attended public and private star parties (with pins awarded to the top three), to writers of reviews for the PAS-times Newsletter and to the weather watcher team. Prizes ranged from a Cosmos Firstscope telescope (thanks, Celestron!) to a 1-year subscription to Astronomy Magazine and books, including one from NASA/JPL about Ikhana, which is a Predator B unmanned aircraft designed to watch for wildfires. (Ikhana is a Choctaw Nation name meaning intelligent.) And there were plenty more, including Moon Watches, and astronomy themed toys and games.

Don announced the elections for next year's officers, giving many of us good reason to focus on our pizza. (Did I say, "Thanks for the pizza, Darren?") Bruce won the Presidency, Terri got Vice President, Don is our Editor, and Mike is the Treasurer. And Alfred is our new Member-At-Large.

After a short break, Sam and Alex presented the results of their month-long weather study to see which online forecast sites might be trusted to accurately predict the cloud cover during an upcoming star party - nobody won...

Last but not least was the final treat, a presentation of video from the PAS Music Jam last December.

And that was the way it went, this year. A good time was had by all, and we all made our way home, happily full of pizza - thanks Darren!

Terri Finch writes: Attending this event was: William & Terri Finch, Don Boyd, Mike Marron, Alex & Diane Vrenios, Darlene Ahlefeld, Jerry Belcher, Earl DeLong, Rodney Fong, Sam Insana, Darren

Johnson, Matt Kohl, Ora Kurland, Howard Moneta, Dick Newell, Eric Steinberg, Kevin Witts, Bette, Ed & Bruce Wurst, Alfred Fantegrossi, Angie Lee and Bob Senzer.

The Agenda for this event went in this order: Bruce opened the meeting and started the Awards Ceremony by reading the Award PAS received from NASA JPL. Then, we presented the Award to the PAS Member who reached the highest number of Messier objects identified during the Messier Marathon at Mike's Home in Carefree on Mar 29 2014, which was Eric Steinberg with 31 Objects! Eric then chose a prize from the prize table, as each Award recipient also did, after they received their awards. The list of prizes and who took which prize, will be listed after this list of the Agenda. Then Bruce read the letter from Night Sky Network which explained what Public Outreach is and why PAS was being recognized for it. We then turned to doing the NSN Awards. NSN Provided 3 pins this year, and so the top three PAS Member who attended the most events, got those three pins.

After all the NSN Awards were given out, we honored the Halloween Costume Contest winner of 2013. Vera Stiesmeyer won with her costume & dance of "Thriller." Sam Insana, her husband, accepted the award for her and chose a prize for her.

We then turned to doing the Elections for which I asked Don to assist, because he knows the rules of elections. So, for the Elections, each officer talked about what their position was, what was expected of them, etc, and then we voted on their positions. The results of the election will be later in this article.

We then took a 5 minutes break and when we returned with more pizza, Sam did a presentation about the weather research he and Alex were working on for the past 35 days, and then Alex did a presentation to follow up on the topic. The gist of what I got from what was said by Sam and Alex about the different weather sites (Intellicast, NOAA, Clear Sky, TWC, W.Underground, Weather Bug, Accuweather, & Skippy Sky) was that, you can't take any one site as being totally accurate. And that the different sites update at different times during the day. Thus, you have to know when the last update was to find out how accurate that prediction will be for the that evening. This is why I really appreciate my Weather Watching Team.

We then enjoyed the 2013 Music Jam video & photo collection put together by William Finch. This was about 2 minutes long. And then, to close out the meeting, we watched about another 25 minutes of PAS 2013 in Photo Review. Many members thanked me for putting together the Year in Photo Review. I think I'll do that again next year, but I'm thinking of having someone assemble it into a powerpoint or other form of slide show, where by the date and what the event was, can be added to the screen as well.

Many thanks to Darren for donating the Pizza from Hungry Howies, to this event. As a group, we donated to the snack fund to reimburse Darren for the pizza. Darren donated it back to the club.

Here's the list of everything that was awarded for and used as a prize for this Awards Ceremony: Sky & Telescope provided the three handouts I made available to - First the award recipients and then to the rest of the attendees. Sky & Telescope also provided the "Binocular Highlights" book, for which William chose as his prize, and the "Pocket Sky Atlas," which Rodney chose for a prize. From Astronomy Magazine, Eric chose the Complimentary Magazine Subscription. From Astronomical Society of the Pacific, "400 Year of the Telescope" book, SOFIA pin (Terri Took), Solar System Lithograph (Sam took), "The Invisible Sky" book (Mike took), and "Space Shuttle" book (Darlene took).

From Celestron "Cosmos Firstscope" taken by Don.

From Oceanside Photo & Telescope "NightWatch" book (which Jerry took). And From Nasa Space Place, 5 copies of "Ikhana", one of which Darren took and Alex took one, too.

I also had extra prizes left over from previous Awards Ceremonies of 2 Moon Watches (Sam took one, Eric took the other), 3 Star Magic Spring Slinkies, 2 X-zylo Boomerang Toys (Ora took for her cats), and 1 Echo Bot (which Matt took). I included 4 ISS Stickers, one of which Don took. And we had 2 donated books by Alex Vrenios which were "Messier Objects: A Beginner's Guide" and "Handbook for Visual Meteor Observations."

Now, who got the awards and for what reason?

I had 5 Weather Watchers this year, who kept me notified of how the weather

PAS Awards Ceremony Apr 24

was going to affect an event: Don Boyd, Alex Vrenios, Eric Steinberg, Sam Insana, and William Finch. Thank you Weather Watchers!

PAS had 6 Members do presentations for the club throughout 2013: Mike Marron did 4 Lectures at his home, Earl DeLong did 2 presentations, Chet Schuler did 1, Eric Steinberg did 1, Chris Johnson did 1 and Alex Vrenios did one. Thank you for your presentations!

Public Outreach: What did this include this year? So, if you attended a Bookmans Telescope Workshop or a PVCC Telescope Workshop, a public Lecture at Mike's home, a CTCA event, an Estrella Observatory or Scout event, and the Indoor/Outdoor event in September 2013, you were added to the list of possibly receiving an award this year. We also had School events, Astronomy Day at Mike's, Black Mountain Campus, ASU West, Thunderbird Park hosted by SAC, A Meteor Shower event any Private event that the public was at, the Grand Canyon star party, or doing Rancho Gabriela Career Day event. All of these type of events were considered in the collection of Public Outreach. The top 3 attendees of these type of events received a pin this year, and anyone doing 3+ events in 2013, received a certificate. If you received a certificate, it also listed the other achievements you did for the year.

++ So, here is the list from top to bottom, of who received an award and for how many Public Outreach events: 39 William Finch, 39 Terri Finch, 37 Don Boyd, 19 Earl DeLong, 19 Mike Marron, 17 Rodney Fong, 13 Eric Steinberg, 12 Sam Insana, 11 Rick Cunningham, 10 Darren Johnson, 10 Joe Collins, 10 Renee Collins, 7 Ora Kurland, 7 Yves Klein, 6 Darlene Ahlefeld, 5 Bob Ewing, 5 Terry Dancer, 5 Jenny Weitz, 3 Frank Insana, and 3 Jerry Belcher.

No separate awards were given for this list, but recognition was announced for Reviews, which are also considered Public Outreach, to these members. The number of Reviews & Articles written for the Newsletter is listed as well: 79 Terri Finch, 13 Leah Sapir, 12 Don Boyd, 10 Joe Collins, 8 Earl DeLong, 7 Alex Vrenios, 5 Eric Steinberg, 4 Darlene Ahlefeld, 3 Bob Ewing, 3 Sam Insana, 1 Yves Klein, 1 Matt Kohl, and 1 Rick Cunningham.

And now we get to who received an award this year from the Night Sky Network and what they did to get it: This is in alpha-

betical order. The order the prizes were awarded is listed above with the Public Outreach list marked with ++ above: Darlene Ahlefeld attended 6 public outreach events and 4 Newsletter submissions. Jerry Belcher attended 3 public outreach events and providing the Dart Game Trophies. Don Boyd (received an NSN Pin) attended 37 public outreach events and 12 Newsletter submissions.

Joe Collins for CTCA Liaison & 10 reviews. Renee Collins for 10 Public Outreach events at CTCA. Rick Cunningham attended 11 public outreach events and 1 Newsletter submissions. Terry Dancer 5 public outreach events attended. Earl DeLong attended 19 public outreach events and 8 Newsletter submissions and 2 Presentations to the club. Bob Ewing attended 5 public outreach events and 3 Newsletter submissions. Terri Finch (received an NSN Pin) attended 39 public outreach events and 79 Newsletter submissions. William Finch (received an NSN Pin) attended 39 public outreach events. Rodney Fong attended 17 public outreach events. Frank Insana attended 3 public outreach events. Sam Insana attended 11 public outreach events and 3 Newsletter submissions. Darren Johnson attended 10 public outreach events and for providing the Pizza for this event. Yves Klein attended 7 public outreach events and 1 Newsletter submissions. Matt Kohl attended 1 public outreach event and 1 Newsletter submission but the real thanks is for all the work on proof reading the PAStimes Newsletter every month. Ora Kurland attended 7 public outreach events. Mike Marron attended 19 public outreach events and 4 Summer Lectures at your home. Leah Sapir attended 13 Newsletter submissions. Eric Steinberg attended 13 public outreach events and 5 Newsletter submissions. Alex Vrenios attended 7 Newsletter submissions & 4 presentations to the Club. Jenny Weitz attended 5 public outreach events and for supporting PAS!

Those of you who have received an award, may pick it up from me at a later time. Find out which event I'm attending, remind me to bring it with me. Only those who attended the event were able to pick a prize from the prize table (must be present to pick a prize).

We hope to see many more PAS members receive awards for the 2104 year. So, be at those events. Sign up, make sure you send me reviews, get the info to me I need, and

let's be awarded for all the support you give PAS!

Thanks to everyone who helped at this event. Matt was my sign in and bouncer for the evening. This was a closed event to PAS Members only. Many thanks to Darren for the wonderful pizza! Many thanks to those who brought extra cameras in case mine died (it has happened before).

Many thanks to everyone who provided photos of events through the year, in which I can put the 2014 photos into the 2014 Year in Photo Review for next year. Send your photos my way and hopefully I will also get all of them into the Photo Gallery over the summer months. That is my goal, out of my computer and into the Photo Gallery for everyone to enjoy. So, if you have photos from 2013 or 2014 you haven't sent my way, start doing it now. Or trade me a burned CD of photos, if you have a lot to give me. Would love to get your photos into the Gallery over this summer!

Many thanks to everyone who attended. Online, I had put a list of the prizes for this Awards Ceremony. I will be updating that list with who chose what prizes and starting the list for the 2014 Awards Ceremony. Many thanks and Congratulations to everyone receiving an award!



Attendance was awesome! Thank you all for coming. Photo by Matt Kohl.



Attendance at the Awards Ceremony was a good amount of PAS Members. WE all enjoyed the Ceremony & many thanks to Darren for the pizza. Photo by Terri Finch.

Archway North Phoenix School May 2

By Terri, Event Coordinator

Attending at this event was William & Terri Finch, Rodney Fong, Frank Insana, Mike Marron, Alex & Diane Vrenios, and Kevin Witts. Our contact was Erin Tabor. Many thanks to Erin and her teacher helpers who provided Dominoes Pizza & water to the Telescope Team.

It was a very hot event. I think the high predicted was to be 98. Event was from 6pm to 9pm, in which part of that was while the Sun was setting. Sunset was 7pm. So, from 6pm to 7pm Alex with Diane's help, set up his Solar Scope. William took out our new PST and set it up. I directed traffic and reminded the kids not to touch the telescopes. This was a very busy, large turnout, event! But Wow, it was really exhausting and fun!

So, before we started, Kevin set up a full table with a really nice table cloth across it with a collection of model rockets and other things that fly in the air, like the Wright Brothers Plane. I asked Rodney to keep an eye on the Rocket table, and Kevin had to chase his planets around. He had set up his Solar System and if it wasn't the winds catching his planets and making them roll away, then it was the kids playing with the planets. Luckily, he gathered them all up, and was able to account for everything by the time he left that night. Many thanks to Kevin for an exciting time at this event!

I shot some photos that i will add to the PAS Photos Gallery for this event. Alex & Diane had brought their camera and shot a few photos too, of the event, and I will include them in the Gallery as well, for all to enjoy!

So, the event got underway.

So, Kevin had his C8. I had my C8 but I didn't do Solar because the kids were spending too much time touching the scopes in the daylight while we were trying to do Solar, so I didn't set up my scope for that. I waited until dusk to set up my scope for the evening. Frank had attended and was helping Mike. Mike told me a few days later, over the phone that if it weren't for Frank being there, being an extra set of eyes, one girl would have stolen one of Mike's meteorites. Thank you so much, Frank, for helping at this event.

While the sun was setting, and while i had a chance, in between chasing kids from touching our scopes, I did Q & A. I gave away 2 posters and a bunch of Lithographs.

My box is getting thin. I will have to refill. After the sun went down, i found the lithographs of the sun. Grrr. I had it planned to give those away since the Sun was what we were having fun with at that moment.

So, the Sun set, and we had Alex on his scope, in which I think he did Jupiter and the Moon for most of the night. I did the Moon & Jupiter, opposite of what he showed. William took Saturn when it became available, as it had to rise up over the nearby mountain. Alex also took Saturn by then. I did a Red Moon, around the time of Saturn. I have no idea what Kevin showed that night.

Many thanks to Erin for inviting us and for the Pizza. Food is always helpful to get us motivated. Many thanks to Kevin and Rodney for sticking around to help William and i haul our stuff up the hill. So, we were set up in a field. It was probably about 10 feet below the parking lot area, no maybe 15 or 20 feet lower in elevation. We could walk down the concrete and grass bleachers that were built into the side of the hill, or we could go down the grassy area, which was about 30 feet out of the way to do so, but easier on my knees. We got all the stuff down to the bottom of the hill and set up.

Then, when it was time to leave at 9pm, we had a crowd join us that just finished a baseball game. So, we hung around until about 9:20 to show them Saturn. Then Kevin and Rodney helped, after packing up everything else, William and I to get everything back up to the edge of the hill where i could pull up the car to load. Kevin kept saying that he doesn't know how I can fit all that in the SATurn Vue. I told him I Tetris-ize it. And i did.

The rocket display was a big hit, but i did ask Kevin to bring less rockets next time so he can keep track of all of it. The Solar System worked well and what we did while Kevin was doing his Solar System was set up our solar scope. Rodney and Frank were invaluable - priceless- so much appreciated!

It was a fun event and the people were so nice. I'm looking forward to another

event like this, same location, but i asked the teacher to be sure to go over the NO Touching the Telescope rules. Oh and we did have a group of kids that didn't believe me when i said i would take away their ball if they can't keep away from the telescopes with it. They were playing kickball and it was getting rather uncomfortable with the ball flying over towards the telescopes, too close for comfort. I told them that twice, and said, one more time and you lose your ball. They kept it away on that third time. Overall, it was an awesome event. Attendance, i'd say was about 350. It was fun and we were so exhausted due to the hill.



Kevin Witts set up a great display at Archway school with his model rockets. Photo by Terri Finch.



Alex, showing the Sun through his safely filtered scope to the attendees of this event. Photo by Diane
(Continued on 23)



A Glorious Gravitational Lens

By Dr. Ethan Siegel

As we look at the universe on larger and larger scales, from stars to galaxies to groups to the largest galaxy clusters, we become able to perceive objects that are significantly farther away. But as we consider these larger classes of objects, they don't merely emit increased amounts of light, but they *also* contain increased amounts of **mass**. Under the best of circumstances, these gravitational clumps can open up a window to the distant universe well beyond what any astronomer could hope to see otherwise.

The oldest style of telescope is the refractor, where light from an arbitrarily distant source is passed through a converging lens. The incoming light rays—initially spread over a large area—are brought together at a point on the opposite side of the lens, with light rays from significantly closer sources bent in characteristic ways as well. While the universe doesn't consist of large optical lenses, **mass itself** is capable of bending light in accord with Einstein's

theory of General Relativity, and acts as a *gravitational* lens!

The first prediction that real-life galaxy clusters would behave as such lenses came from Fritz Zwicky in 1937. These foreground masses would lead to multiple images and distorted arcs of the same lensed background object, all of which would be magnified as well. It wasn't until 1979, however, that this process was confirmed with the observation of the Twin Quasar: QSO 0957+561. Gravitational lensing requires a serendipitous alignment of a massive foreground galaxy cluster with a background galaxy (or cluster) in the right location to be seen by an observer at our location, but the universe is kind enough to provide us with many such examples of this good fortune, including one accessible to astrophotographers with 11" scopes and larger: Abell 2218.

Located in the Constellation of Draco at position (J2000): R.A. 16h 35m 54s, Dec. +66° 13' 00" (about 2° North of the star 18

Draconis), Abell 2218 is an extremely massive cluster of about 10,000 galaxies located 2 billion light years away, but it's *also* located quite close to the zenith for northern hemisphere observers, making it a great target for deep-sky astrophotography. Multiple images and sweeping arcs abound between magnitudes 17 and 20, and include galaxies at a variety of redshifts ranging from $z=0.7$ all the way up to $z=2.5$, with farther ones at even fainter magnitudes unveiled by Hubble. For those looking for an astronomical challenge this summer, take a shot at Abell 2218, a cluster responsible for perhaps the most glorious gravitational lens visible from Earth!

Learn about current efforts to study gravitational lensing using NASA facilities: <http://www.nasa.gov/press/2014/january/nasas-fermi-makes-first-gamma-ray-study-of-a-gravitational-lens/>

Kids can learn about gravity at NASA's Space Place: <http://spaceplace.nasa.gov/what-is-gravity/>



Abel 2218. Image credit: NASA, ESA, and Johan Richard (Caltech). Acknowledgement: Davide de Martin & James Long (ESA/Hubble)

Desert Botanical Gardens Apr 26

(Continued from page 5)

was either there in person or in constant contact by cell phone for any questions we had. The fundraising event by DBG was apparently very successful, having over 600 people attending. Apparently our Star Tours star party was a big success. I know that the 6 of us who worked the DBG star party would like to do more DBG star parties in the future. Thank you Terri for letting the 6 of us, Alex, Kevin, Earl, Howard, Rick and Me do the job this year. Sam Insana PAS Star Tours Member

Amanda Ellis writes: Sam, Good morning! I hope the rest of your weekend went well. I wanted to thank you for a great job at Dinner on the Desert. There has been nothing but positive feedback about the telescope experience! I appreciate you taking time out of your busy schedule to come and be a part of the event. It has been a pleasure working with you and hope to work with you again in the future. Thank you! Amanda ***

Leprecon40 Star Party in Mesa May 10

By Various Attendees

Attending this event, from PAS, was: Don Boyd, Howard Moneta, Kevin Witts, William & Terri Finch. It was a very windy night.

Howard writes: Leprecon was a blast. My schedule opened up for the night so I raced down to Mesa for this event. It was clear that I had arrived at the right convention due to the number of people wandering around the parking lot wearing costumes. The costumes were great although people wearing big hats tend to have trouble getting close enough to the eyepiece. The conversations were fun because some of these sci-fi fans also had some scientific knowledge.

The sky was very clear which made for a pleasant, although somewhat windy, evening of stargazing. I got to use my old joke over and over again while showing Jupiter. "If you think it is windy here tonight, look at those bands on Jupiter, now those are real windstorms!" People usually laugh at that one.

Brought out the large binoculars for moon viewing. The kids loved the moon. Someone asked a girl if she could see the face on the moon. She said that she saw the face and named about 20 other shapes on the surface. Great party, everyone had fun.

Terri writes: Leprecon was an interesting, exhausting day. It was held at the Marriott Hotel & Convention Center in Mesa. It started with William & I leaving home later than planned. We got to Leprecon about 10pm and checked in. While Checking in, I saw PAS's Guest Speaker David Williams from ASU, walking by. We chatted for a bit. He was headed to his room to do a presentation. So, William & I got our day passes and our contact at this convention was Tony Padegimas. We asked him to take us to the location we would be setting up the telescopes. He did that and then we proceeded to see the rest of David's talk "Mars to Curiosity." Well done, David. Since David was up again for another topic "Outer Solar System" we stuck around for most of that presentation. We then went out to our vehicle to have some lunch, which David

was also doing with his wife who was dressed in an awesome Belly Dance outfit.

After lunch, we walked around the convention and finished it in about 1 hour or less. We made our way back to that same room to find a discussion about Habitable Moons, that David was also attending (not presenting). Shortly after that presentation ended, we had to go find Kevin Witts, who was just arriving. So, we caught up to Kevin, got him his day pass, then let him go see the convention while we sat in a "Space Colonization MicroSymposium which was 4 people, one of which was David. That was very interesting. Kevin caught up to us there. After that symposium, Kevin, William and I made our way to the roof to discuss set up of the scopes. And then we each went our own way to go find dinner (This was an extremely fast day).

During Dinner, I got a text from Howard in which he wanted to join the star party. I got permission from Tony to have anyone extra I wanted to attend, so long as they didn't need a day pass. About this same time, Don was arriving and taking his scope upstairs. I waited out front with a luggage cart for Howard to arrive so that I could help him get to where we were setting up.

When we were all set up, working from closest to the building, away... we had a table with my planet tablecloth with William's needed accessories and a chair next to it with Q&A and handouts. Then we had William set up with his 10" scope. Next was Don, with his 8" scope. Kevin was next in line (heading South on the balcony) with his 8" scope, and then Howard took the end of the line of scopes against the railing with his 8" and binoculars on a tripod.

It was a super windy night. One time, while setting up, a box got away from William. So, he had to run downstairs to get it. We were on the second floor balcony. If the wind had blown just a little harder, the box would have went in the pool. Fighting the wind all night, with people viewing Saturn, Jupiter and the Moon, was dizzying through the scopes but the public loved it, and I handed out, I estimate 40+ Q&A and PAS

Bookmarks, and I estimate we saw about 75 public that night. We were scheduled to set up at 6:30, for which we already had people gathering on the deck while we were setting up, and start at 7:30 and go to 10pm. But when 10pm came, we still had people there so we stayed setup until 10:30pm.

Many thanks to my Telescope Team. They were very AWESOME in putting up with the wind and sharing the night sky with the public. I have the most awesome Telescope Team! Many thanks to Tony for having us at the event. And thanks to David for the fine presentations that I hope he does for PAS in the future. We have David scheduled for the November 2014 PAS Meeting.

Tony writes: Everyone, As much as I desire to thank each one of you individually, I am afraid that this one letter will have to suffice. You can readily suppose the reasons.

Now that the chaos is behind us, and we have some small perspective, I would like to share my opinion that programming for Leprecon 40 was a success. This is all because of your efforts, and in many cases in spite of my failures.

Out of 72 participants (not including committee or staff) only six failed to show up, and all of those took pains to inform me as soon as they were able. Many panels were crowded because I anticipated a much higher drop rate.

Of the 120+ scheduled panels or events, over 100 actually happened in one form or another. The most common complaint (once you remove registration) was a wish for more attendees. We will take that as a good sign. Many have complimented how informed and generous the panelists were.

So thank you, thank you, thank you, each and every one of you your part in this lunatic endeavor.

Good luck to all of you in every little thing, and I am sure that most of us will see each other around somewhere. Warm and sincere regards, Tony Padegimas, Outgoing Program Director - Lep40 ***

Bookmans Telescope Workshop May 11

By Terri, Event Coordinator

Many thanks to the 6 PAS members who attended this event, even though there were no public RSVP's. We had a few people swing by, glance at the sign I pointed to that says it is a Telescope Workshop, and walk away, but no public actually attended the event. We think the lack of RSVP might have been because it was Mother's Day.

However, the 6 PAS Members who did, discussed a lot of cool things. Attendance was Bob Senzer, Rodney Fong, Don Boyd, Terri & William Finch, & Howard Moneta. The topics ranged quite a bit, but one of them had to do with setting up a star party on May 23 for the 'predicted to be 100 meteors/hour' Meteor Shower. We discussed it, I emailed Mike about it, and he

and I set that up within the next week. We also discussed presentations for the MOM on May 15. We talked about telescope accessories. It was a productive meeting, even without the public attendance. Many thanks to the 6 PAS Members who attended and for the great conversation. See you at the next Bookmans Telescope Workshop on June 8. ***

Movie Night "Apollo 13" June 21

By Terri, Event Coordinator

It was an awesome night, with clear skies, for a movie at Eric's home. Eric Steinberg and Ora Kurland were perfect hosts for this event. William and I arrived about ½ hour early. Snacks were already being organized. We brought the bottled water, plates and napkins. Eric provided popcorn, chips, salsa, nuts and a few other

tasty items made by Ora, such as Chocolate Cheese Cake and a really delicious muffin.

Additional PAS Members at this event were Don Boyd and Howard Moneta. It was a small group but it was a great movie. Just over 2 hours long. After the movie, we took a short break and then watched an hour long segment on how they gathered the data for this movie, and how they made some of the

scenes. Very interesting. We concluded the movie & extras about 10:30. We talked for a bit longer and left about 11pm. Many thanks to Eric and Ora for hosting this event. They have a wonderful projection TV that makes this type of party a great success. Eric & I will be scheduling the next Movie Night event. ***

Arts & Smarts Day at Bookmans May 24

By Rick Cunningham

This event was a daytime community event sponsored by Bookman's. The primary focus of my involvement this day was solar observation and information decimation. William had his PST and I had my 8" Dobsonian with a full glass solar filter in the parking lot. Other PAS members manned information tables IN THE SHADE by Bookman's entrance. The day was warm and \ started generally clear. By event's conclusion heavy clouds had moved in from the west as forecasted, forcing the cancellation of the evening viewing. Fortunately for solar viewing the mid afternoon cloud cover was not a hindrance.

Indeed, the clouds offered some interesting variations to the solar observation. As the day was warm (upper 90's) only about fifty people ventured over to our telescopes on the blacktop parking lot to view the sun. However, there appeared to be a steady flow of folks at the information tables. Most people were fascinated with the always changing sunspots through my telescope. William's PST showed the mass coronal ejections which to me is more fascinating than the sun spots. A few were afraid to use the telescopes to view the sun believing permanent irreversible damage to their eyes would occur. No amount of education would dissuade their fears. The young, as

always were eager. We even had a Phoenix police officer take a turn at the eyepiece.

After the Bookman's event concluded the seven PAS members present went to Carlos O'Brian's and enjoyed a pleasant meal while re-hydrating.

Terri writes: In attendance at this event, from PAS, was William & Terri Finch with William's Scope, handouts & Q&A, Rick Cunningham, Howard Moneta, Rodney Fong, Bob Senzer and Mike Marron with meteorites.

This day started out as a 2 event day: Solar from 2pm to 4pm and then a Sidewalk Astronomy Event for the evening from 7:30 to 10pm. However, the clouds were not cooperative and the evening event was canceled. But the Solar event was awesome!

Many thanks to Katy Spratt for having us at this fine event.

We set up so that Terri & the Handout/Q & A table were in the shade of the building. Mike set up next to the PAS Table. The Solar Scopes set up out in the parking lot. We had Rick with his 8" and a solar filter and William with our new PST. It was a very hot day to be outdoors, in the sun.

Many, many, many thanks to my Team Members Howard & Rodney. They kept everything going smoothly, taking turns running to the Solar Scopes and then returning to the shade with info. And Howard... wow, thank you so much for providing cold drinks for the TEAM and the chairs. William and I had forgotten the chairs, and Howard came to the rescue. Mike also brought his newest chair which was like an umbrella chair but had a built in cooler. He said he got it at Walgreens. But his chair was also very appreciated. For the next Solar Event at Bookmans, as there will be more... we must remember our chairs! ***



Mike with his Meteorites at Bookmans Solar Event. He always draws a crowd. Photo by Terri Finch.

Mike's Birthday Party May 23

By Alex Vrenios

Review of Mike's birthday/meteor party, May 23, 2014: The peak time for meteor activity was predicted to begin at 11pm Friday and continue for about two hours. The Internet was full of chatter about how good this was going to be, with estimates ranging from 200 to 400 meteors per hour. That's over 6 meteors per minute! A "worthy" shower offers between one and two per minute so I was really pumped.

We started setting up at dusk. There was one telescope and a set of high-powered binoculars on a balance-arm stand. I brought my comfortable but teetering, fold-up swivel chair and a pair of 7x50 binoculars. We all agreed on how lucky Mike was to live in a place with such a beautiful dark sky.

While we all waited for the show to begin, Sam logged a Herschel Object and a comet, and he gave us all a look. I think together we spotted nine satellites. We even saw a few stray meteors (unrelated to the radiant) between 8 and 10pm. I saw three, one of which was bright enough to qualify as a bolide. Together the group claimed about seven meteors, in total. By 10 pm a bank of clouds was getting close to the

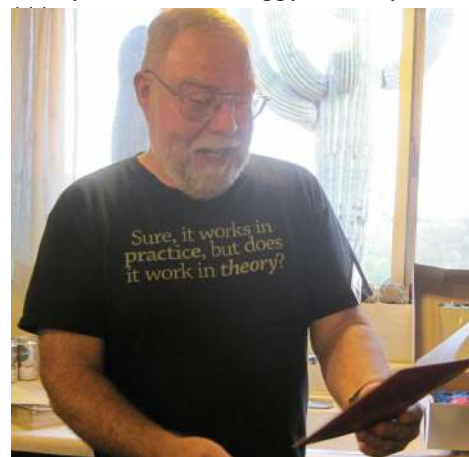
northern horizon, the best viewing area of the sky. By 10:30 that area was almost completely closed off. Reports the next day confirmed that this shower was not the memorable experience we were expecting.

We packed up our stuff and headed indoors to warm up, and discuss Dr. Who and other less notable sci-fi moguls. After a slice or two of leftover pizza I headed home.

Terri writes: In attendance at this event was Don Boyd, William & Terri Finch, Sam Insana, Alfred Fantegrossi, Rodney Fong, Kevin Witts, Kevin Harcey, Alex Vrenios, Bob Senzer, Howard MOneta, Ed, Bette & Bruce Wurst & Angie Lee.

There was lots of pizza at this event. The food was good. Thank you all, for bringing interesting foods to enjoy. PAS paid for the Birthday Cake for Mike, which William and I picked up at Sams Club on the way to Mike's place. We ate and ate, and then we had the birthday cake about 7pm. We sang Happy Birthday & Mike opened his presents. Photos of this event will be in the PAS Photo Gallery soon. Many funny Birthday cards were shared. A few scopes were set up outdoors, as it was a

No Moon night.. As far as Meteors, I saw a total of 1 all night. Several other members saw more meteors. When it got too cold outdoors, we moved indoors and put on a movie. Many of us stayed in the kitchen and chatted at length about Doctor Who and other Sci-fi and TV shows. Many thanks to Mike for hosting this fun event. We look forward to Mike having many more Birthday's so that we can continue to enjoy the dark skies of his location & him, for many more years to come. Happy Birthday Mike!



Mike, the Birthday Boy, reading one of his Birthday Cards at his Birthday party.

Photo by Terri Finch

Memorial Day 2014 at Antenna's

By Eric Steinberg

May 26: While the weather looked good for a Memorial Day observing session, the difficult part this time was finding others willing to brave the heat. In the end, Dick Newell decided to go and we headed out there, arriving around 6:20, meeting just one other observer on-site also. In fact, it was 102 degrees at the time -- but it's a dry heat!! Setup was thirsty work, but when the sun set, it cooled off to a balmy, windless, bug-less, and steady night. Unfortunately, Dick seemed to have some technology issues with his scope and bailed out in the wee hours, but I was able to get through

another 44 Herschel objects - only 30 left to go. Other objects looking great were M13, M11, M51, M101, Saturn and M57. M57 was particularly sharp and actually showed a hint of the central star - the first time I've been able to resolve that. Pride of place for the night however went to the spectacular Omega Centauri which though it was only at 9 degrees altitude, was quite crisp. Using a 14mm 100 FOV eyepiece at 193X felt like swimming in stars! Hope to see you out there.

Dick writes: I went out to the "Antennas" dark site on May 26th. It is definitely out in the middle of nowhere, but I had no problem finding it thanks to Eric's great directions. Seeing was extraordinary and I highly recommend going out there if you have the chance. It is 120 miles from East Mesa where I live, so it took about 1.5 hours to get there. There is a truck stop just west of there which has a lot of light, but is blocked fairly well with a ridge between it and the site. I plan on going to this site frequently, as it's the best that I've seen so far.***

Construction at Black Mountain Campus

By Jenny Weitz

Construction on a new building at Black Mountain is slated to start this October and will most likely begin by digging up our favorite corner of the BM parking lot. As a result there will also be a reduction in the already tight parking on site so blocking off a different area for our star parties may

not be possible. Loretta and I will keep you posted on the state of the parking lot, but I am going to reserve g147 on the main campus on the dates of the star parties next school year as a default site. The good news is that when we make it through the construction and the Black Mountain building

is completed for Fall 2015 it will include a dedicated astronomy pad with power :-). The astronomy pad will be roughly to the West of the new building once construction is complete. ***

Grand Canyon Star Party June 25

By Sam Insana, PAS Member

The Annual Grand Canyon Star Party ran from Saturday June 21st through Saturday June 28th at both the North and South rims of the Grand Canyon.

Sam, Vera and Frank Insana attended the South rim star party Wednesday, June 25th. There was a slide show and lecture at 8pm in the Visitor Center every night showing various topics like astrophotography, differences in telescopes, the night sky, etc... The Tucson Astronomy Club was in charge at the South Rim. This year all the lights were turned off in all the parking lots and all the buildings so that it was a very dark sky, about Bortle 3. There were about 60 telescopes of all shapes and sizes. The public was able to view dozens of deep sky objects from 9 pm until 11 pm. Dennis

Young of Sedona was there with his 28 inch Dobsonian, and there were other amateur astronomers who brought their telescopes from all over the country like Boston, Texas, Pennsylvania, San Diego, etc... The temperature was about 70 degrees, and skies were clear, but there was a wind of about 5-10 mph. The milky way looked incredible.

If anyone wants to attend next year, it will be in June of 2015. Info can be obtained on the internet. If you volunteer to set up your scope for 4 nights or more, you get free park admission and free camping at Mather campground. If you just want to attend to look through the many telescopes, I suggest the South Rim star party since it is closer to Phoenix, has many more tele-

scopes, and is better viewing. The North rim last year was on the balcony of the North rim lodge and had many white lights on, a building and trees in the way and would not be worth the trip to volunteer your services in my opinion. However, if you want to set up your telescope in a Bortle 1 area, you go 20 miles north of the North rim in a meadow and you get the most spectacular telescope viewing of your life. During the day you can see the many great views of the North rim, and in the early evening you can attend astronomy slide shows and lectures in the North rim lodge. Saguaro Astronomy club is in charge of the North rim activities. Contact either club after January 1, 2015 if you wish to volunteer. ***

Desert Botanical Gardens Star Party May 30

By Luana Vargas & Attendees

Thank you for the Astronomy at the Garden program held last week. We had great attendance and I appreciate that you changed the format of the program to include some time spend inside providing an introduction. I wanted to share one of the feedbacks we received from our survey of the program:

Astronomy at the Garden 5/30/14: Would suggest that they spend a bit more time at the beginning setting the stage. Especially since we got outside too early to see anything with the telescopes. For the next program, I suggest adjusting it to provide more time in the classroom or having a later start so that it's darker when they head outside. Probably a combination of both strategies would work best. Thank you, Luana Vargas Program Director, Adult Education.

Terri writes: In attendance at the Adult DBG event this time from PAS was Terri Finch with an 8" scope, William Finch with a 10" Scope, Mike Marron with his Meteorites, and Bruce Wurst with his 20" scope. We got there and DBG had so much going on, parking was interesting. We were finally put in the Bus Loading Zone, where we usually park for these events. We unloaded, set up, and William & I made our way to the room to do the indoor presentation. We were on time, everything was go-

ing well. William did his presentation which was to last 10 minutes but actually only lasted about 5 minutes. His presentation is on how Telescopes Work. I then did my presentation which starts with how to read a planisphere and goes into what we will be seeing that night and what to look for on the various objects (Jupiter's Banding & Moons, as an example). I also only took 5 minutes of the 10 minutes I was given. So, by 7:10pm we were headed outdoors. The group walked faster than I could hobble along, as I was limping this day. We got out to the group and it was still daylight. So, we directed everyone over to Mike and he talked about Meteorites for about 10 minutes while we hunted down the Moon. I had forgotten the Moon was going to be visible, so it was a nice surprise to see the 1 day old Moon in the evening sky. We all found the Moon eventually, and stayed on it for a while. Then Jupiter peeked out, and we all went to Jupiter. As Saturn became visible, I suggested to the group that I would stay on the Moon and let them move to Saturn and Jupiter. Eventually it got dark enough to show some good stuff, but that wasn't until about 8:15pm. So, about that time, we looked for and found the Star Gate Cluster, Sombrero Galaxy, Hercules Globular Cluster. Everyone seemed to enjoy the evening. About 8:45 people were leaving

already, even though the event was scheduled from 7pm to 9pm. And that's when we found the really cool stuff.

During the event, I did a sky tour, where by i had the group sit down, and i showed them around the evening sky. Due to the light pollution, it was more difficult to find the constellation stars. Around closing time, I saw something cool. Looking West, the two main stars of Gemini were peeking like eyes over the mountain to that direction. And Jupiter was positioned just perfect to look like 1 tooth in a grin that was hidden behind the mountain.

Many thanks to Angelica Ellis for having us at this event. Mike was exhausted as was William and I, by the end of this event. Bruce seemed to be doing ok. It was just so hot out, and the 3 of us had done Black Mountain Campus the night before. Many thanks to my Telescope Team! Next time we do the Adult event, I'm asking for 1.5 hours before hand for set up. It was way too rushed to get to the room, after setting up my scope, and then thinking through what I wanted to say. Also, I may make a powerpoint presentation to guide me into what I want to cover. I think I did ok with the presentation and the organization of it, but I did leave some things out that I wish I had told them about, which would have added a few more minutes to my presentation. ***

Black Mountain Campus Star Party May 29

By Various attendees at this event

Bob Ewing writes: Wonderful evening except for being Thursday, which of course means CLOUDS at Black Mtn. Ha! Nice to visit with all you folks.

Eric Steinberg writes: On Thursday, May 29th PAS held a public star party at the PVCC Black Mountain Campus. Around nine scopes and Mike were in attendance, as was a modest but appreciative group from both campuses as well as the public. Weather was initially clear but got progressively cloudier through the evening. Naturally, shortly before the campus closed at 10:00, the skies cleared completely. Never-

theless, views of the visible planets and some DSO's made the event a success from the public point of view. With my 22" scope I showed Jupiter, Mars, Saturn, M13, M92, M11, M4, M5, M10 and comet C 2012 K - though some of those object were only seen by PAS members. Overall a successful, if slightly disappointing night.

Terri writes: In attendance, from PAS, at this event was: Mike Marron, Sam Insana, Howard Moneta, Rodney Fong, Don Boyd, Alfred Fantegrossi, Earl DeLong, William & Terri Finch, Eric Steinberg, Bob Senzer and Bob Ewing. The weather was

hot, sticky, and the clouds weren't super cooperative.

Bob Senzer assisted Mike with his meteorites. William & I worked one scope while doing Q & A. Wililam had his 10" Dob and Terri brought handouts. We had about 150 public in attendance. It was a very good event, enjoyed by all. By the end of the evening, the sky was clearing. Many thanks to my Telescope Team. They kept the crowd busy with many wonderful objects to view. We look forward to the next BMC star party on Sept 18, 2014. See you there! ***

Comicon Star Party June 5, 6, & 7

By Various Attendees

Here's the list of attendees for this event: June 5th had Mike Marron, Kevin Witts, Sam Insana & Frank Insana. June 6 had Mike Marron, Sam Insana, and Howard Moneta. June 7 had Mike Marron, Matt Kohl, William & Terri Finch, Howard Moneta & Kevin Witts. Location of this event was a Phoenix Comicon at the Phoenix Convention Center.

Sam writes: PAS was asked to show the night sky at Comicon 2014. Our club was mentioned in the Comicon Program at page 18. On Thursday night, June 5th, Mike, Kevin, Frank and Sam attended. We set up near 3rd Street and Washington in Downtown Phoenix. The parking was kind of a chore because of all the crowds. At first there were hundreds of bees in our area, but they went away as soon as the sun went down. We set up next to a Mermaid exhibit featuring girls in costume in a pool. Hundreds of people stopped by to look through our scopes. Frank showed the moon through his small refractor, I showed Jupiter and Saturn with my 8 inch newtonian and Kevin showed Jupiter and Saturn with his 8 inch SCT. We showed Mars on request but it wasn't impressive. Mike had his meteorite display which again was a hit. About 10% of the people were in costume, such as batman, superman, spider man, darth vader, harlequin, tinker bell, etc..... Most of the other people had pop culture tee shirts on. Most were thrilled looking at the objects through telescopes. But a few didn't bother... saying they had seen the planets before. There was a very bright street light right next to us, so we couldn't show anything else. We gave away dozens of PAS

handouts that had info on PAS. Some of the crowd were very interested in our club. Being geeks and nerds, they either had telescopes or wanted to buy one soon.

Friday night June 6th was similar except Howard replaced Kevin, and Frank had to go back to the Grand Canyon. There were a few more people and the traffic was a lot worse due to a Diamondbacks game next door. The police were nice, letting us go through barricades to get to our loading and unloading spots. Several police and security guards enjoyed viewing through our telescopes. It was hot out but there were no clouds and no wind. Friday night I put a 2x barlow on my 4.7 mm eyepiece showing a large crater on the moon at 520 power. It looked great.

We were given passes to attend the Comicon events during the day. I attended Friday and Saturday and saw hundred of great costumes, hundreds of exhibits featuring comic books, action figures, tee shirts, paintings, legos, and many other pop culture merchandise. There were dozens of panels and workshops. My favorite was the Adam West, Julie Newmar, Burt Ward panel. About 5000 people attended that one. The 3 actors were very sharp and funny and entertaining. Other panels I attended included Dr Who history, the Big Bang (physics, not the tv show), Astrobiology, Alien language, pop culture planetarium showing where in the night sky famous characters came from (for example they pointed out the constellation where Spock came from), sci fi short film presentations, a fashion show of cosplay, and where we go next in our solar system.

I recommend Comicon to those of you who have never had the chance.

Terri writes: What an exhausting event. But, it was well worth it. It started with getting to Comicon. We left the house, and met up with Sam at his home where Mike was still asleep when we arrived. We picked up our Day Passes to Comicon from Sam. Thank you, Sam for getting them for us. Then we drove down, Matt, William and I, to Comicon, found a parking spot and had a super day.

We first spent a lot of time in the Exhibit Hall where there were kiosks of many things for sale, and to see. We then went up to a panel called "It Started in a Junk Yard" which was all about Doctor Who. We enjoyed that for an hour from 10:30 to 11:30. Then we went back to the Exhibit Hall for more walking, looking, shopping. We ran across Sam in the Exhibit Hall. Then, we headed over to see John Barrowman for an hour. After John's panel, we walked back to the car and had lunch. We then went back into the Exhibit Hall until about 3:30 and then tried to get into the Nathan Fillion panel, but it was full, so we went back to the Exhibit Hall. We ran across Mike outside, as he was dropping off stuff in his car, and picking up meteorites to talk about with people. We finished out the Exhibit Hall and went out to the car for dinner. We packed our lunch and dinner. Then we loaded stuff up and the three of us walked it over to the observing area which was outside the South Convention Center, at the S. W. Corner.

Matt was a great help during the star party. He took care of the handouts, giving

Comicon Star Party June 5, 6, & 7

everyone a copy. We had brought probably 800 copies with us and gave out nearly half. Plus, during the event, I did Q & A and probably gave out 80 prizes.

William was set up with his 10" dob. Kevin brought his 8" and Howard had his 8", and under the light, with an audience most of the time, was Mike with his meteorites.

We set up between 6:30 and 7:30, but those who were ready, started once they were set up. We were supposed to end at 10pm but the crowd kept coming, so we stayed there until 11:30. We showed the Moon, Mars, and Saturn all night. Jupiter was available until about 10pm when it went behind the building. I'd estimate we had 500+ people visit our observing area. It was so many people, and many return people. Before it was dark, we had a bunch of people stop by and we told them that once it was dark, it was worth coming back for another view, so they did. We had several people say, in passing on their way to someplace more important, that they have seen Saturn and Jupiter before. ??? We had many people ask us about the Riparian location that EVAC has events at, and we clarified we weren't that group. I had 2 teachers stop by who are planning to set up events with us during next school year. It was really an amazing, exhausting day.

The temperature for the evening was hot. I'm glad the sun was down when we were setting up. William and I drank like fish all day, and still it wasn't enough. The sky was perfectly clear all night. It really was a perfect viewing night. And the planets, moon, etc all cooperated and looked great. So, at 10pm, when we were supposed to set up for this event, I asked Kevin and Howard if they wanted to close up for the

night. Kevin and Howard both said they were having a great time and wouldn't mind staying set up. So, we did until 11:30pm.

For next year.... All PAS members who want Day passes, must sign up to do that night's star party. The day you do the star party is the day they give you a pass for. Also, next year, we are going to get out in the walk way more, and spread out more... we had plenty of room to keep Mike under the light with his meteorites, and have the 3+ scopes move into the walk way so people have to see us and come look through our scopes. We were also considering a canopy to block the light for one of the scopes, and on the canopy we can put up the PAS sign. I'd like to do more than one day at Comicon, next year and make it to more panels and other activities.

I wish to thank Greg Cooling for inviting us to Comicon this year. I never got to meet Greg but I hope he had a great time, too. I'd like to thank Disa Holly, a previous member of PAS, for suggesting we get involved with Comicon. I'd also like to thank the Telescope Team for all the hard

work they did all three days. Let's think about doing this event again next year. ***



Photo shows Howard standing by his scope on the left, Matt doing handouts & Kevin behind him with his scope, in front is William pointing at something with the Laser Pointer and his scope. We had a continuous crowd from set up at 6:30 to 11:30pm. It was an amazing event.

Photo by Terri.



Mike at Comicon Star Party with a crowd on June 7! Photo by Terri.

Bookmans Telescope Workshop June 8

By Terri, Event Coordinator

Many thanks to my wonderful Telescope Team at Bookmans on this fine Sunday afternoon. We had William & Terri Finch, Bob Senzer, Howard Moneta, Matt Kohl & Kevin Harcey and Rodney Fong. Attending this event was Susan & her husband John Walsh and their 12 year old daughter Catherine, who got the telescope for her birthday. We had originally gotten 2 RSVP's, but the 2nd one canceled earlier on Sunday morning due to health reasons.

The afternoon started with the 5 PAS Members enjoying some great conversation

around the PAS Handout table. A gentleman wandered over to talk about Doctor Who with us, since I mentioned Comicon. He stuck around for about 1/2 hour before the Walsh's arrived. We had John set up the scope and then everyone realized the eyepieces were left at home. So, Rodney and Howard gave some great advice on how to align the scope. Susan took notes.

It was a very fun afternoon. After the Walsh's left, the topic turned towards improving handouts for PAS, updating the "Who are we" page, and a few other things

we can do to draw more people into PAS. Howard, William and I were sore from doing the Comicon star party the night before, and exhausted. Many thanks to Bookmans for allowing us to use their location to hold this event. We look forward to the next Bookmans Telescope Workshop on July 13.

Susan writes: We enjoyed the event. Our questions were answered. John did align the scope when we got home. We look forward to the lecture and stargazing on the 26th of July. ***

CTCA/PAS SkyTour Event 6-11-2014

Attendees: Mike Marron, Sam Insana, Don Boyd, Joe Collins.

Patient/Caregiver Attendees: n= 6 (including 2 who did not sign in).

Sam Insana PAS member: On June 11th, Joe, Don, and I showed the night sky to people at the Cancer Treatment Center of America. We showed Jupiter, Mars, Saturn and the Moon. I also showed M13 the Hercules globular cluster. Don tried astrophotography through his new telescope, getting good shots of the moon and some trial shots of Saturn, using prime focus photography. We also saw some strange lights to the south like the Phoenix lights. They were orange and round and were not flares since there was no smoke showing when viewed through binoculars. However, Joe said he thinks he knows what they are, some kind of magnesium lights on military planes or helicopters or drones. Mike showed his meteorites to the people. During a 30 minute break between dining and the star party, I played the baby grand piano in the lobby.

Don Boyd PAS Member: I looked at Jupiter, Saturn, the moon, the Beehive, Alcor and Mizar, the Stargate cluster, and I tried to look at M5 the butterfly cluster but could not find it, which brings me to the problem I had with my scope that night. The "Gotos" were very erratic all night; some were right on and some were way off. I tried redoing the alignment and it was still off. Then I checked the gears on the RA motor and I noticed that one of the gears would turn almost 1/8 of a turn before the shaft turned...so I have a set screw that is loose,

an easy fix but very annoying. Earlier in the night, I had to adjust the Dec motor as somehow it had slipped and the gears were not making contact. I had to loosen the mounting bolt and move it closer. Finally got it resolved.

Joe Collins PAS member and Cancer-Fighter: We cancelled solar viewing due to high temps. It was a great night for lunar gazing, two days before the "strawberry Moon" on Friday the 13th (an event that won't repeat itself for 35 years). I showed patients the Moon, Jupiter, Saturn, Albireo, Mizar, Polaris, M13 and several other lesser known binary stars. There were some strange lights to the south which were probably part of a military exercise. I tried my new Celestron StarSense for the first time, but did not get it collimated to my OTA in time for customers who arrived early and the display was hard to read, so I will try the procedures again before our next event.

I will remark that even though my eyes couldn't see them (because it wasn't really dark yet and we had some slight overcast at the very beginning of the event), the StarSense camera picked out 45 stars in the small section of sky it was pointing at., so I know it is going to work well once I get it figured out. Our guest participation numbers were down tonight, so we are going to start and end an hour earlier next time the Moon is up for observation (Good suggestion Sam!) and I am going to give out busi-

ness cards for the event to patients and care-givers we pass by the day of the event.

Mike Marron PAS member: People enjoyed seeing the meteorites as typical. I had fun showing my meteorites to patients and caregivers both in the downstairs Café and upstairs on the 5th floor.

We want to thank CTCA and Amanda Montelongo the Patients Events Coordinator in particular for support of our events. We have been invited to continue our events at CTCA for the next 12 months starting next event July 9th! ***



Ward Traditional Academy May 2

Written by Terri, Event Coordinator

In attendance at this event was a fantastic Telescope Team of Sam Insana, Rick Cunningham, and Don Boyd. The contact at the school was Jason Blackstock, and this school is located in Tempe.

Sam writes: On May 2nd, PAS gave a star party at Ward middle school in Tempe. Don, Rick, and Sam attended. We showed sunspots and solar flares from 6 pm until 6:45 pm. A flock of about 100 Ibis's flew over near sunset, an unexpected treat. Jason, our school contact, is a 5th and 7th grade science teacher. He provided us with pizza and drinks and had a large popcorn machine to feed the crowd. At 7 pm we started showing the moon. Then at 7:30 pm we showed Jupiter. At 8 pm we were show-

ing Mars. At 8:30 pm there were a lot of oooh's and aaaah's when we showed Saturn. There was too much sky glow from the valley to show galaxies or nebulas or globular clusters, but we did show some double stars and open star clusters. Don showed us an Iridium flare. There were about 100 teachers, parents, and school kids in attendance. Several of the students were very sharp in their astronomy knowledge and one 7th grade student wanted info on our telescope workshops, PAS lectures and future star parties. I gave him our PAS website info. It was a nice evening with no wind and about 10% cloud cover.

Jason writes: Thank you for your email. The event was perfect. All three

astronomers were great and worked very well with the kids. We ended up having 80% of 5th grade attend the night, most with their parents who were all very impressed. Overall, I would estimate the total around 80 - 100 people. The night was a huge success! Thank you so much for your help with all the details and the organization. As soon as we have a check cut, I will make sure it gets to you. Please let me know if there are any other details that we need to work out as well.

I want to plan another event for next year, perhaps in early April, which we may expand to some more grades. Thank you for the great night! ***

Telescope Workshop at PVCC May 22

By Terri, Event Coordinator

In attendance at this Telescope Workshop, from PAS was: William & Terri Finch, Don Boyd, Mike Marron, Howard Moneta, Bob Senzer, Rodney Fong and Jenny Weitz. Many thanks to Jenny for taking the time to allow us to do this event. There were no classes happening at PVCC, so she took her time to help us have this event and it was a great success.

Peter attended with his son Matt. They brought their Celestron scope with them. Howard, Don and William helped Pete. Mike kept Matt busy with meteorites. The telescope started out indoors. Then, the group moved outdoors to try the tracking. After some discussion and testing the

equipment, it was decided the scope needed some help. I texted President Bruce and asked him if he was able to assist. He said yes, so Pete and Bruce got together with the scope and now it works!

Many thanks to the Telescope Team & Jenny. We look forward to the next Telescope Workshop at PVCC on Sept 25, 2014. See you there! Photos of this event will be in the PAS Photo Gallery, soon. ***



Mike always has a crowd around him. He is like a magnet that draws in the interested people. Photo by Kevin Adams.

Black Mountain Campus Apr 21

(Continued from 7)

One of his students had a camera, and I spent some time chatting with her about what she was doing and suggesting other ways to do it, based on my past photo experience. She was photographing the night sky. I didn't get to see the photos but Bob and his class were chatting with the girl about it for quite some time.

Rodney's scope had a nice image of Jupiter and then Saturn. Jodi & I talked about her views of Jupiter through her scope. When I got to Howard's scope, I was amazed at the line up of the Moons. So, we had Jupiter and all 4 moons were to one side. But what was really cool was that the three moons closest to Jupiter were in a line, just like Orion's Belt stars. Then the 4th moon was further away from the planet. That was really awesome to see. I have never seen that configuration of the moons. Very awesome.

Some really cool objects that I was told were viewed but that I didn't see on this night, were: the Ghost of Jupiter, the Stargate Cluster, & Sombrero Galaxy.

Around 9pm, I wandered over to Mike to see what he was doing and Bob Senzer & Mike were keeping the attention of a group. They were talking about a bunch of different things having to do with meteorites, science, cosmology and more. Many thanks to Bob for joining the TEAM to-night.

Many thanks to my Telescope Team! They all did a fantastic job of the night and

I really appreciate having them there. We look forward to the next several Black Mountain Campus Star Parties. Here are the dates in 2014: May 29, Sept 18, Oct 23, Nov 20, and Dec 11. We hope to see everyone at all these upcoming fun, public events!

Loretta writes: Yes I was so glad to see the turnout. Thanks so much for organizing this for us.

Kevin writes: Weather was great for

group to nighttime viewing through Binoculars and they returned several times to reuse them, looking at the Pleiades, Orion's Nebula and other celestial objects.

Through my scope I was able to view Jupiter, Saturn and Mars, The Perseus Double Cluster, the Stargate, Arrow Cluster, #3 Cluster and some Messier objects including the Cigar galaxy and the Ghost of Jupiter. A personal highlight included seeing a moon transit across Jupiter through Eric's scope



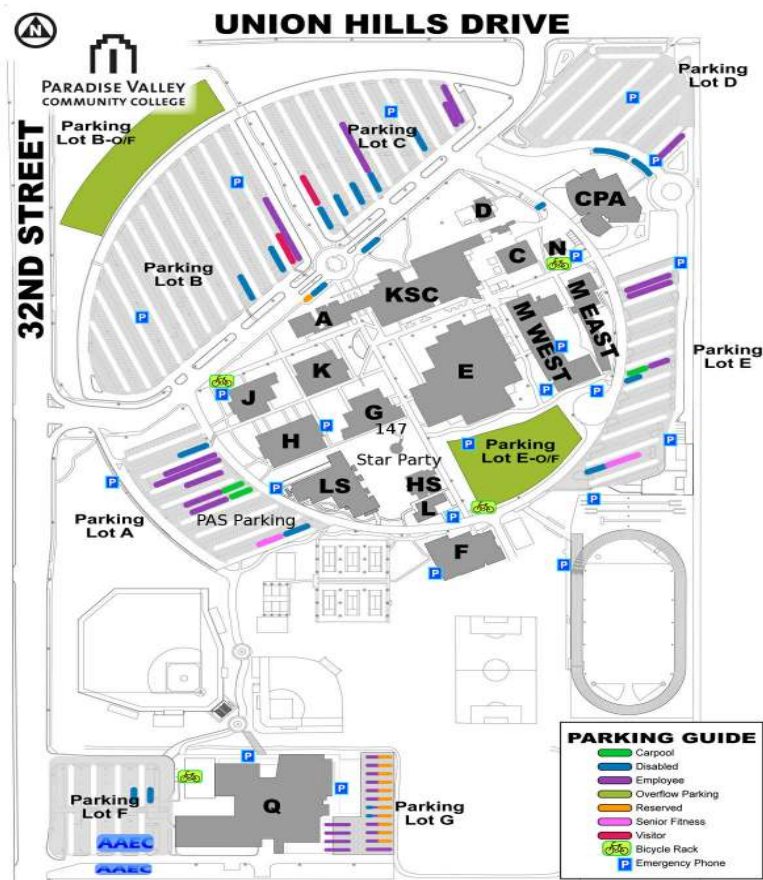
Telescope set up at BMC in April. We had a great time!

the star party as the skies were extremely clear, there was minimal wind and the atmosphere appeared to be very stable. I had approximately 40 people look through my scope including a large group of Astronomy students. The students were required to view 10 objects and list the type of telescope, the magnification, and the sky coordinates of each object. I introduced a small

and also watching the ISS cross the sky. At one point, a student was looking at Mars and had the added treat of seeing a meteor pass in front of it through the scope. Jupiter was particularly good tonight and Eric's view of Mars was pretty spectacular as well.***

Map of PVCC Main Location

18401 N. 32nd Street | Phoenix, AZ 85032



Map of PVCC Black Mountain

34250 N. 60th Street | Scottsdale, AZ 85266



Dynamite Montessori School May 2

By Terri, Event Coordinator

In attendance at this event was Eric Steinberg. Our contact at this school was Paula Fabian Leach.

Eric writes: On May 2, 2014, PAS member Eric Steinberg provided a single-scope star party for the Dynamite Montessori school in North Phoenix. Setup began about 6:15, during which the school provided a delicious home-cooked dinner; observ-

ing began with the moon around 7:30. As usual, attendees were amazed at the views through the 22" and expressed their appreciation. Other objects shown through the evening were Jupiter, Mars, M37, M3 and Saturn. There were a number of other activities for the group of perhaps 40 adults and children, from which they broke away through the evening to enjoy the views of the sky. The Montessori kids are a spirited

bunch, but overall very pleasant - other than a bit of an ongoing battle with lots of white LED flashlights, the evening was an enjoyable success, with Paula and Gene who run the school being extremely gracious throughout.

Paula writes: The event went very well! Eric and his scope are amazing! Thank you for connecting us. ***

Archway North Phoenix School May 2

(Continued from page 13)

Alex writes: I could not have planned a better evening for a star party. A group of three hundred school children and their parents got to see sunspots and solar flares until sunset, when we switched to a beautiful crescent moon. When it got a little darker, we added Mars and Jupiter, with all four of its Galilean moons in full view. Later still, Saturn and Titan appeared over the mountain. Lines of twenty or more people at each scope kept us busy the whole night. ***



The Kids helped Kevin set up his Solar System. Photo by Terri Finch.

Desert Botanical Garden Kids Event May 31

By Terri, Event Coordinator

In attendance this night, from PAS was Don Boyd, Terri & William Finch & Mike Marron. We had about 18 public attending this event. Emily Morris was my contact and we had a wonderful assistant there, who did the photos last year, too.

The evening began with Emily setting up her Kid activities. This event was from 7:30 to 9:30, so when we started, it wasn't dark yet, so we gathered everyone who was on time to the event, over by Mike and let Mike take the crowd for the first 20 minutes or so. Then, as we were able to capture objects, we showed the Moon first. Then Jupiter became available. Saturn peeked out next and then it got dark enough to do a sky tour. I did the sky tour for about 10 minutes. We continued to show the planets until it was dark enough to find M13, Sombbrero Galaxy (which was super faint) and Ring Nebula, and the Star Gate cluster. Since we had just done this location the night before, for the Adult Event, we had a good idea what was available to show during the Kid Event.

The temperature was hot, but the skies were clear. It was a pleasant night, and went by really fast. Many thanks to Emily who worked with me to make this event go

smoothly. After the sky tour, while we were still showing some great items, Emily did a book reading. The kids enjoyed that activity.

They also enjoyed the various activities Emily supplied, one of which was making craters on the moon using cornstarch and plastic balls to play the part of the asteroids hitting the Moon. There was cornstarch everywhere, but it was fun! Many thanks to Emily for having us at this event! We hope to have more events like this one in the near future. And many thanks to my Telescope Team. ***



Mike with his Meteorites at DBG. Photo by Terri Finch.



Telescope Set up at DBG. Don, William & my scopes by the Boojam Tree. Photo by Terri.

2014 PAS GUEST SPEAKER LINE-UP

By Terri, Event Coordinator Events@pasaz.org

Do you have an idea for a Guest Speaker? Email me the details. Here's the scheduled Guest Speakers at PAS:

<http://www.pasaz.org/index.php?pageid=meetings>

Sept 4: Members Night - Anyone Sign up to do a mini presentation.

Oct 2: Mark Murison: Topic: TBA

Nov 6: David Williams: Topic "Mars: From Mariner to Curiosity & Beyond: A History of Mars Exploration."

PAS is incorporated in the state of Arizona as a non-profit, scientific and educational organization. Our newsletter PAStimes is published monthly from September to May and distributed by US PS and the Internet. All Issues are available for download on our website www.pasaz.org Ads for astronomy equipment are provided as a courtesy to sellers and buyers, and do not constitute any endorsement by PAS. All Photos by Don Boyd unless otherwise credited. All articles and photos are copyright their authors or PAStimes.

What's Up For July

By Rod Sutter, PAS Past President

Name	Date	Rise	Set
Mercury	07-15-14	04:04	18:57
Venus	07-15-14	03:34	17:45
Mars	07-15-14	12:49	00:02
Jupiter	07-15-14	06:03	20:01
Saturn	07-15-14	14:29	00:18
Uranus	07-15-14	23:35	12:14
Neptune	07-15-14	21:52	09:09
Pluto	07-15-14	18:38	04:53

All Times Arizona Time

July 15 2014

Sunrise: 05:29

Sunset: 19:38

New: June 27

Q1: July 5

Full: July 12

Q3: July 18

